

Basefold in the List Decoding Regime^{*}

Ulrich Haböck

StarkWare Industries
`ulrich@starkware.co`

Abstract. In this document we discuss the soundness of the Basefold multilinear polynomial commitment scheme [ZCF23] applied to Reed-Solomon codes, and run with proximity parameters up to the Johnson list decoding bound. Our security analysis relies on a generalization of the celebrated correlated agreement theorem from [BCI⁺20] to linear *subspaces* of Reed-Solomon codes, which turns out a by-product of the Guruswami-Sudan decoder analysis from [BCI⁺20]. We further highlight a non-linear variant of the subspace theorem, which is flexible enough to apply to Basefold-like protocols such [Dia24b] a row-batched version of FRI-Binius [DP24], and WHIR [ACFY24].

^{*} This is a revision of the preprint [Hab24], which was written while the author was at Polygon Labs. The present version clarifies the soundness analysis in terms of round-by-round soundness, and discusses soundness of WHIR as a polynomial commitment scheme.

Table of Contents

1	Introduction	1
2	Preliminaries	3
2.1	Interactive oracle proofs	3
2.2	Round-by-round soundness	4
2.3	Polynomials	5
2.4	Sumcheck protocol	6
3	Basefold for Reed-Solomon codes	7
3.1	The protocol	9
3.2	Sketch of soundness	12
4	Correlated agreement with constraints	14
4.1	For linear subspaces	16
4.2	...and beyond	18
5	Soundness in the oracle model	19
5.1	Batching step	21
5.2	Folding steps	22
5.3	Query phase	24
5.4	Skipping commitments	24
6	Generalizations	24
6.1	More expressive inner products	25
6.2	Other FFTs	27
6.3	FRI-Binius	28
6.4	WHIR	31

1 Introduction

A *polynomial commitment scheme (PCS)* is a cryptographic primitive, in which one party, the prover, is given the possibility for providing *evaluation proofs* on previously committed polynomials, i.e. cryptographic proofs of their value at arbitrary query points. Polynomial commitment schemes are a key component of modern argument systems, which essentially are polynomial *interactive oracle proofs*, that are then turned into a cryptographic argument by instantiating the ideal PCS (the oracle) with the concrete scheme.¹ Polynomial commitment schemes come in two flavors, serving either univariate polynomials, or multivariate, most importantly *multilinear*, polynomials. Their design rely on various mathematical structures, for example pairings [KZG10, PST13], general elliptic curves [BCC⁺16, BBB⁺18], lattices [AFLN23, CMNW24, NS24], and groups of unknown order [BFS20, SB23]. An alternative line of work uses error-correcting

¹ We are aware that this simplistic view falls short of many technical aspects of code based constructions, in particular in the list decoding regime. However, for the sake of brevity we keep with the term PCS.

codes, foremost Reed-Solomon codes, for example [AHIV17, BBHR18, BGKS20, BLNR22, DP23] or more recently FRI-Binius [DP24] and the proof of proximity for circle codes [HLP24]. Univariate and multilinear schemes are closely related, and there are general transformations for turning an univariate into a multilinear one, and vice versa, see [ZXZS20, BCHO22, CBBZ22, KT23] and [PH23].

Basefold [ZCF23] is a multilinear polynomial commitment scheme built from *foldable codes*. Foldable codes, as introduced by Zeilberger, Chen and Fisch, are error-correcting codes over arbitrary finite fields, which possess an encoding procedure similar to the Fast Fourier Transform (FFT), thereby admitting a proximity test similar to the *Fast Reed Solomon Code Interactive Proof of Proximity (FRI)* [BBHR18]. On very high level, the protocol underlies the following mechanics. Multilinear polynomials are mapped to code words so that linear combinations of their “even” and “odd” parts (in the generalized sense) translate to partial evaluation of the multilinear. With this parallelism in mind, the value at a given query is then proven via the multivariate sumcheck protocol [LFKN92], round-wise intertwined with the proof of proximity, sharing the same verifier challenges.

Although random foldable codes have acceptable metrics [ZCF23], the most important use cases remain Reed-Solomon codes in all facets, over binary fields [DP24], over elliptic curves [BCKL21, BCKL22] or the circle curve [HLP24], which is due to their tighter and more extensive soundness analysis. This brings us to the main purpose of this paper: We specialize the soundness analysis from [ZCF23] to the case of Reed-Solomon codes, and extend it to the *list decoding regime*, allowing proximity parameters up to the Guruswami-Sudan-Johnson bound $1 - \sqrt{\rho}$, where ρ is the rate of the code. Concretely, we make use of the celebrated correlated agreement theorem from [BCI⁺20] to argue soundness in a round-by-round manner, similar to the analysis of FRI therein. In the unique decoding regime, that is for proximity parameters up to $(1 - \rho)/2$, the round-by-round analysis faces no particular difficulties. However, beyond the unique decoding radius, things are slightly more delicate. Correlated agreement alone is not sufficient to link a “successful” FRI folding with the sumcheck specialization, and we need to take a closer look at the list decoder analysis from [BCI⁺20] in order to generalize the statement accordingly. This leads to two strengthenings of the correlated agreement theorem:

- One for linear *subspaces* of Reed-Solomon codes, sufficient for Basefold and FRI-Binius [DP24] with its special structured evaluation inner product; and
- a non-linear generalization, which is useful in the analysis of Basefold-like protocols that serve more expressive inner products, for example WHIR [ACFY24] and the FRI-Binius variants [Dia24a, Dia24b].

The round-wise errors in the soundness analysis are essentially the errors imposed by the (strengthened) correlated agreement theorem, yielding, up to negligible differences the same proof sizes as FRI as an univariate PCS, without the overhead of the aforementioned generic univariate-to-multivariate transformations.

The document is organized as follows. In Section 2 we introduce notation, basic facts on multilinear polynomials, interactive oracle proofs, and the sumcheck protocol. Section 3 recaps the Basefold for Reed-Solomon codes in the simplest setup, over finite fields with a smooth multiplicative subgroup. We sketch the underlying mechanics of our soundness proof, and explain the need of strengthened correlated agreement. In the following Section 4, we then go over the corner pillars of the Guruswami-Sudan list decoder analysis from [BCI⁺20] and derive the above mentioned strengthenings of the correlated agreement theorem. While this part, considered in full depth, is the most challenging, we stress the fact that it can be easily understood on high-level, without going into the algebraic knits and grits of the decoder analysis. Section 5 is the core of the document: We prove round-by-round soundness of Reed-Solomon Basefold in the oracle model, for proximity parameters up to the Johnson bound. The proof is in the spirit of that of FRI in [BCI⁺20] and makes use of the correlated agreement strengthening from Section 4. The remainder of the document, Section 6, is then devoted to variants and generalizations of Basefold: The adaptation of Basefold to other FFTs is outlined in Section 6.2, and Basefold for more expressive inner products is addressed in Section 6.1. FRI-Binius, in its optimized form [Dia24b], is discussed in Section 6.3.

The present document is a revision of the previous preprint [Hab24]. Besides clarifying that the analysis therein is not merely in a round-wise manner, but proves round-by-round soundness [CCH⁺18, CMS19], the present version further adds a soundness analysis of WHIR [ACFY24] in the list decoding regime, Section 6.4.

Acknowledgements. The author would like to thank: Ben E. Diamond and Jim Posen for posing the question addressed in this writeup; Swastik Kopparty for his feedback on Section 4; Giacomo Fenzi for his feedback and for sharing an early version of [ACFY24]; Al Kindi for a thorough proof read; and Ariel Gabizon for pointing out a missing univariate translation of constant multilinear in the centering step of Lemma 1 and 2.

2 Preliminaries

2.1 Interactive oracle proofs

An *interactive proof* [BS84, BM88, GMR89] between two parties is an interactive protocol, in which one party (the *prover*) wishes to convince another party (the *verifier*) upon the validity of a claimed statement. An *interactive oracle proof* (IOP) [BCS16, RRR16], or an interactive oracle proof in the *plain oracle model*, is an information-theoretic model of an interactive proof in which the prover is able to commit data strings via *oracles*. Oracles serve the ideal functionality of a binding and hiding commitment, and allow partial revelations (local *openings*) at arbitrary positions of the string, *queried* by the verifier. The oracles we use

will be of the form $f \in F^D$, where D is some well-defined subset of a finite field F , and the queries will be points $x \in D$ selected by the verifier.

The interactive oracle proofs discussed in this document will be as follows. We presume the prover has committed to functions $f_{0,1}, \dots, f_{0,M_0} \in F^D$, together with some auxiliary plain text data aux_0 , altogether forming the prover message

$$m_0 = \{(f_{0,1}, \dots, f_{0,M}), aux_0\}.$$

The interactive oracle proof of m_0 belonging to a subset of certain claimed properties,

$$m_0 \in \mathcal{R}_0,$$

is comprised of a fixed number of $r \geq 1$ rounds, where in each round $i = 1, \dots, r$ the verifier sends a random challenge $\rho_i \leftarrow F$ drawn from a specified subset of F , and the prover responds by committing to a new set of functions, besides some auxiliary plain text data,

$$m_i = \{(f_{i,1}, \dots, f_{i,M_i}), aux_i\}.$$

We shall use the common terms that the prover *sends* the functions to the verifier, and keep in mind that the verifier is only given oracle access to them. After these rounds, the verifier queries the functions it has received at a specified number of random positions (from their respective domain of definition), and uses the local openings in order to decide whether to accept or reject, by returning a verdict v to the prover. We thus think of the protocol transcript, the messages exchanged between the prover and the verifier, arranged as

$$m_0; \underbrace{\rho_1, m_1}_{\text{round 1}}, \underbrace{\rho_2, m_2}_{\text{round 2}}, \dots, \underbrace{\rho_r, m_r}_{\text{round } r}, v,$$

where m_0 is the presumed prover message, (ρ_i, m_i) are verifier challenge and prover answer of the rounds $i = 1, \dots, r$, and v is the final decision by the verifier. This convention slightly deviates from the one often used in literature, which considers m_0 as part of the first round. We do so as it is the natural choice in the context of this work: Interactive oracle proofs are proofs of proximity, with the target set $\mathcal{R}_0$ claiming the existence of proximate polynomials $p_1(X), \dots, p_M(X)$, such that

$$d((f_{0,1}, \dots, f_{0,M}), (p_1, \dots, p_M)) < \theta,$$

with an appropriate notion of distance, and these polynomials satisfy certain additional properties.

2.2 Round-by-round soundness

We view an interactive oracle proof as a chain of randomized reductions

$$\mathcal{R}_0 \xrightarrow{\rho_1} \mathcal{R}_1 \xrightarrow{\rho_2} \dots \xrightarrow{\rho_r} \mathcal{R}_r,$$

which gradually reduces the claim that $m_0 \in \mathcal{R}_0$ to that the collected prover messages $(m_0, \dots, m_i)$ belong to

$$\mathcal{R}_i = \mathcal{R}_i(\rho_1, \dots, \rho_i),$$

typically a set with a simpler membership check, a simpler “language”. The soundness error of a reduction step $\mathcal{R}_{i-1} \xrightarrow{\rho_i} \mathcal{R}_i$ is the threshold probability under which the desired reverse conclusion,

$$(m_0, \dots, m_{i-1}, m_i) \in \mathcal{R}_i(\rho_1, \dots, \rho_{i-1}, \rho_i) \Rightarrow (m_0, \dots, m_{i-1}) \in \mathcal{R}_{i-1}(\rho_1, \dots, \rho_{i-1}),$$

might not be true. Concretely, if $(m_0, \dots, m_{i-1}) \notin \mathcal{R}_{i-1}(\rho_1, \dots, \rho_{i-1})$ then the probability such that there exists a potential prover message m_i , which extends $(m_0, \dots, m_{i-1})$ to an element of $\mathcal{R}_i$,

$$\Pr_{\rho_i} [\exists m_i \text{ s.t. } (m_0, \dots, m_{i-1}, m_i) \in \mathcal{R}_i(\rho_1, \dots, \rho_{i-1}, \rho_i)] \leq \varepsilon_i,$$

where the bound is independent of the prior messages, that are $(m_0, \dots, m_{i-1})$ and $(\rho_1, \dots, \rho_{i-1})$. In other words, if any (computationally unbounded) prover strategy $\mathcal{A}$ is able to answer the verifier challenge ρ_i with a message m_i that extends $(m_0, \dots, m_{i-1})$ to an element compliant with $\mathcal{R}_i$, with a probability

$$\Pr \left[(m_0, \dots, m_{i-1}, m_i) \in \mathcal{R}_i(\rho_1, \dots, \rho_i) \mid \begin{array}{c} \rho_i \leftarrow \$ F \\ m_i \leftarrow \mathcal{A}(m_0, \rho_1, m_1, \dots, m_{i-1}, \rho_i) \end{array} \right] > \varepsilon_i,$$

then the previous messages $(m_0, \dots, m_{i-1}) \in \mathcal{R}_{i-1}(\rho_1, \dots, \rho_{i-1})$. We stress the fact that our view as a chain of randomized reductions is merely equivalent way to express round-by-round soundness as in [CCH⁺18, CMS19], which uses a (not necessarily efficient) state function to tell whether $(m_0, \dots, m_i) \in \mathcal{R}_i$ or not. The reduction errors $\varepsilon_1, \dots, \varepsilon_r$ from above coincide with the separate round-by-round errors, and

$$\varepsilon = \max_{1 \leq i \leq r} \varepsilon_i$$

is the round-by-round soundness error of the complete protocol. Although not touched in this document, we assume that the reader is aware of the fact that in the random oracle model, interactive oracle proofs (with a public coin verifier) can be compiled into non-interactive arguments via the BCS-transform [BCS16], see also the excellent manuscript [CY24].

2.3 Polynomials.

Let F be a finite field of arbitrary characteristic. We denote the set of univariate polynomials of degree less than 2^n by

$$\mathcal{P}_n = F[X]^{<2^n},$$

for any integer $n \geq 0$. The n -dimensional *Boolean hypercube* over F is the set $H_n = \{0, 1\}^n$, regarded as a subset of F^n , and a multivariate polynomial $P \in$

$F[X_1, \dots, X_n]$ in n variables $X_1, \dots, X_n$ is *multilinear*, if it is at most linear in each of its variables, $\deg_{X_i}(P) \leq 1$ for every i , and thus of the form

$$P(X_1, \dots, X_n) = \sum_{i=(i_1, \dots, i_n) \in \{0,1\}^n} c_i \cdot X_1^{i_1} \cdot \dots \cdot X_n^{i_n},$$

with coefficients $c_i \in F$. For any $\mathbf{x} = (x_1, \dots, x_n) \in F^n$ we write $P(\mathbf{x})$ for the value $P(x_1, \dots, x_n)$. Multilinear polynomials are uniquely determined by their values over the Boolean hypercube, and their coefficients can be computed from the values by a multidimensional FFT, and vice-versa.

The n -dimensional *Lagrange kernel*, also called $eq(\cdot)$ in several works, is the multilinear polynomial

$$L(X_1, \dots, X_n, Y_1, \dots, Y_n) = \prod_{i=1}^n (1 - (X_i + Y_i) + 2 \cdot X_i \cdot Y_i), \quad (1)$$

where we will use the same notation for different dimensions n , without causing confusion. For any $\mathbf{y} = (y_1, \dots, y_n) \in \{0, 1\}^n$ its specialization $L(X_1, \dots, X_n, \mathbf{y}) = L(X_1, \dots, X_n, y_1, \dots, y_n)$ is the unique multilinear from $F[X_1, \dots, X_n]$ which evaluates to 1 at $\mathbf{y}$, and is equal to 0 elsewhere on H_n . This property induces that for arbitrary multilinear $P \in F[X_1, \dots, X_n]$ and $\mathbf{y} \in F^n$, the inner product

$$\langle P, L(\cdot, \mathbf{y}) \rangle = \sum_{\mathbf{x} \in H_n} P(\mathbf{x}) \cdot L(\mathbf{x}, \mathbf{y}) = P(\mathbf{y}), \quad (2)$$

thus yielding the value of P at $\mathbf{y}$. We shall call this inner product the *evaluation inner product*.

2.4 Sumcheck protocol.

The multivariate sumcheck protocol [LFKN92] is an interactive proof for that a multivariate polynomial $G \in F[X_1, \dots, X_n]$ satisfies

$$\sum_{\mathbf{x} \in H_n} G(\mathbf{x}) = s.$$

Typically, the multivariate G is a virtual polynomial in several committed multilinear, and its maximum individual degree $d = \max_i \deg_{X_i}(G)$ is at least 2. Although we shall describe our protocols in a self-contained manner, we quickly sketch the sumcheck protocol, as it is a core ingredient of Basefold. Before the first round the prover claims the refinement polynomial

$$q_0(X) = \sum_{(x_2, \dots, x_n) \in H_{n-1}} G(X, x_2, \dots, x_n),$$

which is of degree at most d . The verifier checks the sanity condition that $s = q_0(0) + q_0(1)$, chooses a random λ_1 , and it asks the prover to prove the specialized claim

$$q_0(\lambda_1) = \sum_{(x_2, \dots, x_n) \in H_{n-1}} G(\lambda_1, x_2, \dots, x_n)$$

instead. The protocol continues in the same manner, letting the prover provide refinement polynomials

$$q_i(X) = \sum_{(x_{i+2}, \dots, x_n) \in H_{n-(i+1)}} G(\lambda_1, \dots, \lambda_i, X, x_{i+2}, \dots, x_n),$$

for $i = 1, \dots, n-2$ as response of the received challenges $\lambda_2, \dots, \lambda_{n-1}$. In the last step the verifier samples $\lambda_n \leftarrow F$, which eventually reduces the sumcheck claim to that

$$q_{n-1}(\lambda_n) = G(\lambda_1, \dots, \lambda_n),$$

a evaluation claim for the multivariate polynomial G at the random point $\lambda = (\lambda_1, \dots, \lambda_n)$, comprised of the random challenges received in the course of the protocol. For a detailed treatment of the sumcheck protocol, and its importance to multivariate proofs in general, we refer to the manuscript [Tha23] and the references therein.

3 Basefold for Reed-Solomon codes

In this section we describe Basefold [ZCF23] for Reed-Solomon codes over a finite field F with characteristic $p > 2$, having a smooth multiplicative group, meaning that

$$G_n = \{x \in F : x^{2^n} = 1\}$$

is a multiplicative subgroup of order $|G_n| = 2^n$, for a given fixed integer $n \geq 1$. For such a smooth group, as well as any of its subgroups and their cosets, interpolation and evaluation of polynomials can be done via the Fast Fourier Transform.

Setup. With the existence of smooth multiplicative subgroups, there is the FRI low degree test [BBHR18] for

$$\mathcal{C}_0 = \text{RS}_{2^n}[F, D] = \{q(x)|_{x \in D} : q(X) \in \mathcal{P}_n\},$$

the Reed-Solomon code generated by the space of polynomials $\mathcal{P}_n = F[X]^{<2^n}$, over any evaluation domain D which is a union of cosets of G_n .² We fix the evaluation domain D , and with it the rate

$$\rho = 2^n/|D|$$

of the code. As FRI, Basefold goes along the chain of projected domains

$$D = D_0 \xrightarrow{\pi} D_1 \xrightarrow{\pi} \dots \xrightarrow{\pi} D_n,$$

² Often D is taken as a (coset of a) larger subgroup than G_n , but this is not necessarily needed to make the protocol work.

where the domains halve under each application of $\pi(x) = x^2$, and their codes

$$\mathcal{C}_k = \text{RS}_{2^{n-k}}[F, D_k] = \{q(x)|_{x \in D_k} : q(X) \in \mathcal{P}_{n-k}\},$$

generated by the space of polynomials of degree less than 2^{n-k} , for $k = 0, \dots, n$. Note that each $\mathcal{C}_k$ has again rate ρ . It is helpful to be familiar with the mechanics of FRI, the round-wise folding of odd and even parts. (However, even without it, the reader should be able to follow the presentation.)

Value-to-coefficient identification. In Reed-Solomon Basefold, a multilinear polynomial $P \in F[X_1, \dots, X_n]$ is committed through its *univariate representation* $p(X) \in \mathcal{P}_n$, defined through its values over the Boolean hypercube $H_n = \{0, 1\}^n$,

$$p(X) = \sum_{i=0}^{2^n-1} P(i_1, \dots, i_n) \cdot X^i, \quad (3)$$

where $(i_1, \dots, i_n) \in \{0, 1\}^n$ are the integer bits of $i = i_1 + i_2 \cdot 2 + \dots + i_n \cdot 2^{n-1}$. Concretely, it is committed as word $f \in \mathcal{C}_0$, by evaluating $p(X)$ on the evaluation domain D . Note that (3) establishes a one-to-one correspondence

$$\Phi_n : F[X_1, \dots, X_n] \longrightarrow \mathcal{P}_n \quad (4)$$

between the space of multilinear polynomials $F[X_1, \dots, X_n]$ and the space $\mathcal{P}_n$ of univariate polynomials of degree less than 2^n . (In the edge case $n = 0$, Φ_0 is the identity mapping between the space constant multilinear polynomials and constant polynomials.) We stress the fact that other identifications, such as coefficients-to-coefficients [ACFY24], work in a similar manner, but come with higher encoding costs for most applications, which typically represent witness data as values over the hypercube.

The main idea. With the identification (3) in mind, the even and odd part $p_0(X), p_1(X)$ of the univariate representation $p(X)$, subject to

$$p(X) = p_0(X^2) + X \cdot p_1(X^2),$$

correspond to the prefixed polynomials $P(0, X_2, \dots, X_n)$ and $P(1, X_2, \dots, X_n)$, and more generally, any linear combination of the form

$$p_{\lambda_1}(X) = (1 - \lambda_1) \cdot p_0(X) + \lambda_1 \cdot p_1(X)$$

corresponds to the specialization $P_{\lambda_1}(X_2, \dots, X_n) = P(\lambda_1, X_2, \dots, X_n)$ of the multilinear polynomial. In other words, *FRI-like folding corresponds to partial evaluation of the multilinear representation*³, and repeating the argument we conclude that, eventually the last oracle of the folding cascade corresponds to the value

$$v = P(\lambda_1, \dots, \lambda_n),$$

where $\lambda_1, \dots, \lambda_n$ are the randomnesses drawn in the course of the protocol. To leverage this observation for custom, non-random, queries $\omega = (\omega_1, \dots, \omega_n) \in F^n$, Basefold considers the evaluation inner product

$$v = \langle L(\omega, \cdot), P(\cdot) \rangle_{H_n} = \sum_{\mathbf{x} \in H_n} L(\omega, \mathbf{x}) \cdot P(\mathbf{x}),$$

and proves it via the multivariate sumcheck protocol *intertwined* with the folding rounds of the FRI proximity test: In each round, both reductions, the one of the sumcheck protocol and the one of FRI, use the *same* randomness.

3.1 The protocol

We slightly deviate from the presentation in [ZCF23] in regards to how the prover provides the sumcheck polynomials

$$q_i(X) = \sum_{\mathbf{x}=(x_{i+2}, \dots, x_n) \in H_{n-(i+1)}} L((\lambda_1, \dots, \lambda_i, X, \mathbf{x}), \omega) \cdot P(\lambda_1, \dots, \lambda_i, X, \mathbf{x}), \quad (5)$$

determined by the round challenges $\lambda_1, \dots, \lambda_i$, where $1 \leq i \leq n-1$. Instead of the quadratic polynomial $q_i(X)$ we let the prover send the *linear* polynomial

$$A_i(X) = \sum_{\mathbf{x}=(x_{i+2}, \dots, x_n) \in H_{n-(i+1)}} L(\mathbf{x}, (\omega_{i+2}, \dots, \omega_n)) \cdot P(\lambda_1, \dots, \lambda_i, X, \mathbf{x}), \quad (6)$$

from which the original polynomial can be reconstructed via

$$q_i(X) = L(\lambda_1, \dots, \lambda_i, \omega_1, \dots, \omega_i) \cdot L(X, \omega_{i+1}) \cdot A_i(X). \quad (7)$$

(The equality is an immediate consequence of the product structure of the Lagrangian.) Although this representation change can be seen as prover optimization [Gru24, DT24], the reason for it is a different one: The linear polynomial allows to relate the folding specialization on the FRI-side of the protocol with the refinement on the sumcheck side, using the correlated agreement theorem for *linear subspaces* of Reed-Solomon codes. See the sketch of soundness in the following section. The concrete protocol, presented as an interactive oracle proof is as follows.

Protocol 1 (Reed-Solomon Basefold [ZCF23]). *Let $P \in F[X_1, \dots, X_n]$ be a multilinear polynomial in $n \geq 1$ variables, $\omega = (\omega_1, \dots, \omega_n)$ be any query from F^n , and $v = P(\omega)$. The prover takes the Reed-Solomon code word $f_0 \in \mathcal{C}_0$ corresponding to the univariate representation of P , computes the linear polynomial $A_0(X)$ as in (6) and shares*

$$f_0, A_0(X),$$

and $s_0 = v$ with the verifier. Then both engage in the following protocol, consisting of a commit phase and a subsequent query phase.

³ This and similar connections have been used in previous constructions, see [BCHO22, CBBZ22, Ham22, KT23] and [PH23].

1. *Commit phase. This is the multivariate sumcheck protocol intertwined with the folding cascade of FRI, using n reduction steps.*

- (a) *In round i , $1 \leq i \leq n-1$, the verifier previously received $f_{i-1} \in F^{D_{i-1}}$, $\Lambda_{i-1}(X)$ from the prover, and knows the value s_{i-1} . It checks that the sumcheck polynomial $q_{i-1}(X)$ determined from $\Lambda_{i-1}(X)$ using (7) satisfies*

$$s_{i-1} = q_{i-1}(0) + q_{i-1}(1),$$

and returns a random scalar $\lambda_i \leftarrow \$ F$ to the prover.

(In the first round $i = 1$, Formula (7) collapses to $q_0(X) = L(X, \omega_1) \cdot \Lambda_0(X)$.)

The prover computes the Reed-Solomon codeword $f_i \in \mathcal{C}_i$ corresponding to $P_{\lambda_1, \dots, \lambda_i} \in F[X_{i+1}, \dots, X_n]$, and its linear polynomial $\Lambda_i(X)$ as in (6), and answers the verifier with

$$f_i, \Lambda_i(X).$$

Both take $s_i = q_{i-1}(\lambda_i)$ for the next round.

(In round $i = n-1$, Definition (6) reduces to the singleton sum $\Lambda_i(X) = P(\lambda_1, \dots, \lambda_{n-1}, X)$.)

- (b) *The last round $i = n$ is as the previous rounds, except that the prover replies on $\lambda_n \leftarrow \$ F$ with the constant $c = P(\lambda_1, \dots, \lambda_n)$ for the constant code word f_n (no Λ_n either), sent in plain to the verifier. Both take $s_n = q_{n-1}(\lambda_n)$.*

The verifier does the final check whether $s_n = L(\omega, \lambda_1, \dots, \lambda_n) \cdot c$, and proceeds with the query phase, which consists of $s \geq 1$ query rounds:

2. *Query Phase. In each of the rounds, the verifier samples a random $x_0 \leftarrow \$ D_0$, and queries the oracles $f_0, \dots, f_{n-1}$ for the values that are needed to check the consistency relations*

$$\begin{aligned} f_i(x_i) &= \text{fold}_{\lambda_i}(f_{i-1})(x_i) \\ &= (1 - \lambda_i) \cdot \frac{f_{i-1}(x_{i-1}) + f_{i-1}(-x_{i-1})}{2} + \lambda_i \cdot \frac{f_{i-1}(x_{i-1}) + f_{i-1}(-x_{i-1})}{2 \cdot x_{i-1}}, \end{aligned}$$

for every $i = 1, \dots, n$, along the projection trace of x_0 , given by $x_i = \pi^i(x_0)$. (f_n is taken as the constant function c .)

If all verifier checks of the protocol, including those of the sumcheck, pass then the verifier accepts. (Otherwise, it rejects.)

Of particular importance is the generalization of Protocol 1 to collections of multilinear, often called “batches”, as correlated agreement across all the committed words plays a crucial role in soundness of the application. We keep with a single common query $\omega \in F^n$ for the entire batch of multilinear. Multi-query proofs are discussed in Section 6.1.

Protocol 2 (Batch Reed-Solomon code Basefold). *The prover shares the Reed-Solomon codewords $g_0, \dots, g_M \in \mathcal{C}_0$ of the multilinear $G_0, \dots, G_M$, together with their evaluation claims $v_0, \dots, v_M$ at $\omega \in F^n$ with the verifier. Then they engage in the following extension of Protocol 1:*

1. *In a preceding round $i = 0$, the verifier sends a random $\lambda_0 \leftarrow \$ F$, and the prover answers with the oracle for*

$$f_0 = \sum_{k=0}^M \lambda_0^k \cdot g_k. \quad (8)$$

Then both prover and verifier engage in Protocol 1 on f_0 and the claim $v_0 = \sum_{k=0}^M \lambda_0^k \cdot v_k$. In addition to the checks in Protocol 1, the verifier also checks that equation (8) holds at every query x from D_0 .

We state round-by-round soundness of Protocol 2, as an interactive oracle proof for the target “language”

$$\mathcal{R} = \left\{ (g_0, \dots, g_M) : \begin{array}{l} \exists p_0, \dots, p_M \in \mathcal{P}_n \text{ s.t.} \\ d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta \\ \wedge \bigwedge_{k=0}^M P_k(\omega_1, \dots, \omega_M) = v_k, \\ \text{where } P_k = \Phi_n^{-1}(p_k) \end{array} \right\}, \quad (9)$$

without specifying the intermediate relations of the protocol as a chain of randomized reductions. The detailed treatment is postponed to Section 5. We assume the proximity parameter θ in the list decoding regime, concretely

$$\theta = 1 - \left(1 + \frac{1}{2m}\right) \cdot \sqrt{\rho},$$

where $m \geq 3$ is the multiplicity parameter of the Guruswami-Sudan list decoder. However we stress the fact that the entire analysis, and in particular Theorem 1 below, carries over verbatim to the unique decoding regime, with the adapted soundness errors for Theorem 3.

Theorem 1 (Basefold soundness). *Protocol 2 as an interactive oracle proof for $\mathcal{R}$ satisfies round-by-round soundness, with round-by-round soundness error*

$$\varepsilon \leq \max \left\{ \varepsilon(\mathcal{C}_0, M, 1, \theta), \frac{1}{|F|} + \varepsilon(\mathcal{C}_1, 1, B_1, \theta), \dots, \frac{1}{|F|} + \varepsilon(\mathcal{C}_n, 1, B_n, \theta), (1 - \theta)^s \right\}. \quad (10)$$

Here, $\varepsilon(\mathcal{C}_i, M_i, B_i, \theta)$ is the soundness error of the weighted correlated agreement theorem, Theorem 3, for subspaces of $\mathcal{C}_i = \text{RS}[F, D_i, 2^{n-i}]$, on a batch of $M_i + 1$ words, using weight denominator bound $B_i = |D|/|D_i| = 2^i$, and proximity parameter θ .

Proof. Follows from Section 5, Lemma 1, Lemma 2, Lemma 3, and that the round-by-round soundness error is the maximum of the roundwise errors. $\square$

Let us quickly outline the components of the soundness error in Theorem 1. The first term $\varepsilon(\mathcal{C}_0, M, 1, \theta)$ is the error for the batching step of the $M + 1$ given words over $D_0 = D$. This is the same error as given by the regular (i.e. non-weighted) correlated agreement theorem for linear subspaces, Theorem 2 below, and we chose the weighted error only for notational convenience. Second, the terms

$$\frac{1}{|F|} + \varepsilon(\mathcal{C}_i, 1, B_i, \theta), \quad i = 1, \dots, n,$$

are the errors for the further, FRI-like reduction steps, when folding $M_i + 1 = 2$ words over the domain D_i , where the additional $1/|F|$ term is a remainder contribution to the sumcheck. One would expect a double as large term, since the sumcheck is quadratic. However, half of it is already covered by the subcode correlated agreement theorem, and the other half can be also dropped when using the improved non-linear variant from Section 4.2. See Lemma ?? in Section 6.1. Finally, the $(1 - \theta)^s$ -term is the soundness error of the query phase.

3.2 Sketch of soundness

The crucial ingredient for proving the soundness of Protocol 2 is the following strengthening of the correlated agreement theorem for Reed-Solomon codes [BCI⁺20]. It allows to relate the specialization on the FRI-side of the protocol with the refinement of the sumcheck claim. A proof of the theorem, as well as its weighted variant Theorem 3, is given in Section 4.1.

Theorem 2 (Correlated agreement for subcodes). *Let F be a finite field of arbitrary characteristic, and $\mathcal{C} = \text{RS}_k[F, D]$ the Reed-Solomon code over F with evaluation domain $D \subseteq K$ and rate $\rho = k/|D|$. Let $\mathcal{C}'$ be a linear subcode of $\mathcal{C}$, generated by a subspace $\mathcal{P}'$ of polynomials from $F[X]^{<k}$. Given a proximity parameter $\theta = 1 - \sqrt{\rho} \cdot (1 + \frac{1}{2m})$, with $m \geq 3$, and words $f_0, f_1, \dots, f_M \in F^D$ for which*

$$\frac{\left| \{z \in F : d(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, \mathcal{C}') < \theta\} \right|}{|F|} > \varepsilon,$$

where

$$\varepsilon = M \cdot \frac{\left(m + \frac{1}{2}\right)^7}{3 \cdot \rho^{\frac{3}{2}}} \cdot \frac{|D|^2}{|F|}, \quad (11)$$

Then there exist polynomials $p_0, p_2, \dots, p_M \in \mathcal{P}'$, and a set $A \subseteq D$ of density $|A|/|D| \geq 1 - \theta$ on which $f_0, f_1, \dots, f_M$ jointly coincide with the values of $p_0, p_1, \dots, p_M$, respectively.

Let us illustrate the role of Theorem 2 in proving the soundness. We do this by means of the first folding round of Protocol 1, for a given proximity parameter θ .⁴ Suppose that a (possibly malicious) prover is able to provide f_0 and Λ_0 , the

⁴ The importance of the theorem in the batching round of Protocol 2 is even easier to explain.

latter of which is sumcheck compliant, $s_0 = q_0(0) + q_0(1)$, and so that with noticeable probability the linear combination

$$f_{\lambda_1} = (1 - \lambda_1) \cdot f_{0,0} + \lambda_1 \cdot f_{0,1}$$

is θ -close to a polynomial $p_{\lambda_1}(X) \in \mathcal{P}_{n-1}$ the multilinear representation $P_{\lambda_1} \in F[X_2, \dots, X_n]$ of which satisfies the sumcheck refinement

$$\langle L((\lambda_1, \cdot), \omega), P_{\lambda_1} \rangle_{H_{n-1}} = q_0(\lambda_1). \quad (12)$$

We wish to conclude that then both $f_{0,0}, f_{0,1}$ agree with polynomials $p_0(X), p_1(X)$ from $\mathcal{P}_{n-1}$ on a joint set $A \subseteq D_1$ of density at least $1 - \theta$, the multilinear representations P_0, P_1 of which are again sumcheck compliant,

$$\langle L((0, \cdot), \omega), P_0 \rangle_{H_{n-1}} = q_0(0), \quad (13)$$

$$\langle L((1, \cdot), \omega), P_1 \rangle_{H_{n-1}} = q_0(1). \quad (14)$$

Once this is given, it is then easy to see that over the preimage $\pi^{-1}(A) \subseteq D_0$, the word f_0 agrees with the combined polynomial $p(X) = p_0(X^2) + X \cdot p_1(X^2)$ from $\mathcal{P}_n$, the multilinear representation of which satisfies the sumcheck claim, since $q_0(0) + q_1(0) = s_0$.

To prove (13) and (14) we reduce the quadratic sumcheck compliance to consistency with the linear polynomial A_0 . By Equation (7), sumcheck compliance (12) translates to

$$L(\lambda_1, \omega_1) \cdot \langle L(\cdot, \omega_2, \dots, \omega_n), P_{\lambda_1} \rangle_{H_{n-1}} = L(\lambda_1, \omega_1) \cdot A_0(\lambda_1),$$

and hence, except for a set of probability $1/|F|$ (for the zero of $L(\cdot, \omega_1)$), we get that

$$\langle L(\cdot, \omega_2, \dots, \omega_m), P_{\lambda_1} \rangle_{H_{n-1}} = A_0(\lambda_1),$$

which, after centering, reads as

$$\langle L(\cdot, \omega_2, \dots, \omega_m), P_{\lambda_1} - A_0(\lambda_1) \rangle_{H_{n-1}} = 0.$$

By linearity, $A_0(\lambda_1) = (1 - \lambda_1) \cdot A_0(0) + \lambda_1 \cdot A_0(1)$, and the constant multilinear in $n - 1$ variables corresponds to multiples of the univariate polynomial $\Phi_{n-1}(1) \in \mathcal{P}_{n-1}$.⁵ We conclude that, with noticeable probability the random linear combination of the centered parts,

$$f'_{\lambda_1} = (1 - \lambda_1) \cdot \underbrace{(f_{0,0} - A_0(0) \cdot \Phi_{n-1}(1))}_{=: f'_{0,0}} + \lambda_1 \cdot \underbrace{(f_{0,1} - A_0(1) \cdot \Phi_{n-1}(1))}_{=: f'_{0,1}},$$

⁵ The concrete form of the univariate is $\Phi_{n-1}(1) = 1 + X + \dots + X^{2^{n-1}-1} = \frac{X^{2^n-1}-1}{X-1}$, a scaled version of the univariate Lagrangian. We would like to thank Ariel Gabizon for pointing out the missing $\Phi_{n-1}(1)$ in a previous version of the write-up.

is θ -close to the centered polynomial $p'_{\lambda_1} = p_{\lambda_1} - \Lambda_0(\lambda_1) \cdot \Phi_{n-1}(1)$, which belongs to the space

$$\mathcal{P}'_{n-1} = \{u(X) \in \mathcal{P}_{n-1} : \langle L(\cdot, \omega_2, \dots, \omega_n), U \rangle_{H_{n-1}} = 0\}.$$

This polynomial space defines a linear subspace $\mathcal{C}'_1$ of the Reed-Solomon code $\mathcal{C}_1$, and by Theorem 2 we are able to conclude the following: Both centered words $f'_{0,0}, f'_{0,1}$ agree with polynomials $p'_0(X), p'_1(X)$ from $\mathcal{P}'_{n-1}$ on a joint set $A \subseteq D_1$ of density $\geq 1 - \theta$. Back in terms of non-centered functions, both even and odd parts $f_{0,0}$ and $f_{0,1}$ agree over A with the respective polynomials

$$\begin{aligned} p_0(X) &= p'_0(X) + \Lambda_0(0) \cdot \Phi_{n-1}(1), \\ p_1(X) &= p'_1(X) + \Lambda_0(1) \cdot \Phi_{n-1}(1), \end{aligned}$$

from $\mathcal{P}_{n-1}$, which moreover are consistent with the sumcheck side of the protocol: Their multilinear representations $P_0, P_1 \in F[X_2, \dots, X_n]$ satisfy

$$\begin{aligned} \langle L(\cdot, \omega_2, \dots, \omega_n), P_0 \rangle_{H_{n-1}} &= \Lambda_0(0), \\ \langle L(\cdot, \omega_2, \dots, \omega_n), P_1 \rangle_{H_{n-1}} &= \Lambda_0(1). \end{aligned}$$

Again by (7) the latter two constraints imply the sumcheck compliance, equation (13) and (14).

The complete argument in Section 5 faces the same technicalities as the soundness proof for FRI in [BCI⁺20, Chapter 8.1]. The analysis keeps track of the folding consistencies by means of consistency measures, and it relies on the weighted variant of the correlated agreement theorem, Theorem 3, to argue soundness of each folding step.

4 Correlated agreement with constraints

In this section we discuss how the correlated agreement theorem for linear subspaces of Reed-Solomon codes, Theorem 2, as well as its non-linear generalization, Section 4.2, are obtained from the Guruswami-Sudan list decoder analysis in [BCI⁺20]. To this end, we give a sufficiently detailed overview of [BCI⁺20, full version, Chapter 5].

Let $F = \mathbb{F}_q$ be a finite field (of arbitrary characteristic), and $\mathcal{C} = \text{RS}_k[F, D]$ be the Reed-Solomon code of rate $\rho = k/|D|$ generated by $\mathcal{P} = F[X]^{<k}$, the set of all polynomials of degree less than $k \geq 1$, using an arbitrary evaluation set $D \subseteq F$ of size $|D|$. We fix a proximity parameter

$$\theta = 1 - \left(1 + \frac{1}{2 \cdot m}\right) \cdot \sqrt{\rho},$$

with integer $m \geq 3$ being the multiplicity parameter for the Guruswami-Sudan list decoder.

Assume that we have given words $f_0, f_1 \in F^D$ so that $S = \{z \in F : d(f_0 + z \cdot f_1, \mathcal{C}) < \theta\}$ is of size

$$|S| > \frac{(m + \frac{1}{2})^7}{3 \cdot \rho^{\frac{3}{2}}} \cdot |D|^2, \quad (15)$$

and let

$$P_z \in \mathcal{P}, z \in S, \quad (16)$$

be an *arbitrary* selection of θ -proximates, satisfying $d(f_0 + z \cdot f_1, P_z) < \theta$ for every $z \in S$. (Here d is the fractional Hamming distance over D . Weighted agreement is discussed later on.) In order to understand list decoding across different z 's, [BCI⁺20] analyze the decoder over the rational function field $K = F(Z)$ in the indeterminate Z .

The proof starts with determining

$$Q(X, Y, Z) \in F[X, Y, Z],$$

a carefully selected polynomial variant of the Guruswami-Sudan interpolant for the K -valued word

$$f_0 + Z \cdot f_1,$$

meaning that $Q(x, f_0(x) + Z \cdot f_1(x), Z) = 0$ in K , with multiplicity at least the given parameter m , for each $x \in D$. (This is Step 1 in [BCI⁺20, Chapter 5].) Let

$$Q(X, Y, Z) = C(X, Z) \cdot \prod_i R_i(X, Y, Z)^{e_i}$$

be its decomposition into irreducible polynomials. The main part of the proof, Step 2 to Step 7 in [BCI⁺20, Chapter 5], is then devoted to showing that one⁶ of the irreducible factors $R_i(X, Y, Z)$ is in fact of the form $Y - P(X, Z)$, with $\deg_X(P) < k$ and $\deg_Z(P) \leq 1$, and hence leads to a solution

$$Y = P(X, Z) = p_0(X) + Z \cdot p_1(X), \quad (17)$$

with $p_0(X), p_1(X) \in \mathcal{P} = F[X]^{<k}$. The proof uses techniques for factorizing bivariate polynomials over finite fields (i.e. computing a power series solution from a simple root via Hensel lifting) carried over to an algebraic extension of K ; a step that requires certain familiarity with algebraic function fields.

Proposition 1 ([BCI⁺20], Proposition 5.5). *Under the above assumptions, there exists a polynomial $P(X, Z) \in F[X, Z]$ of degree $\deg_X P < k$, $\deg_Z P \leq 1$, and so that*

$$|\{z \in S : P(X, z) = P_z(X)\}| > \frac{|S|}{2 \cdot \ell_m},$$

where $\ell_m = (m + \frac{1}{2}) / \sqrt{\rho}$ is the Guruswami-Sudan list size bound for multiplicity parameter $m \geq 3$. In a nutshell, the factor ℓ_m reflects the pigeon-hole principle for at most list size many irreducible

factors, the additional factor 2 is for excluding poles of the sufficiently many coefficients of the power series solution.

⁶ Namely, any factor which covers a sufficiently large fraction of the claimed proximates.

The analysis generalizes to more expressive linear combinations, in particular to the case $f_0 + Z \cdot f_1 + \dots + Z^M \cdot f_M$, with $f_0, \dots, f_M \in F^D$ for any $M \geq 1$, while the larger degree in Z demands the size of the set

$$S = \{z \in F : d(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, \mathcal{P}) < \theta\}$$

being scaled by the degree M ,

$$|S| > M \cdot \frac{(m + \frac{1}{2})^7}{3 \cdot \rho^{\frac{3}{2}}} \cdot |D|^2. \quad (18)$$

This generalization is proven in [BCI⁺20, Section 6.2], and we cite it as a separate proposition.

Proposition 2 ([BCI⁺20], Section 6.2). *Under the above assumptions, there exists a polynomial $P(X, Z) \in F[X, Z]$ of degree $\deg_X P < k$, $\deg_Z P \leq M$, and so that*

$$|\{z \in S : P(X, z) = P_z(X)\}| > \frac{|S|}{2 \cdot \ell_m},$$

where $\ell_m = (m + \frac{1}{2}) / \sqrt{\rho}$ is the Guruswami-Sudan list size bound for multiplicity parameter $m \geq 3$.

Proposition 1 and respectively Proposition 2 are the core results from which correlated agreement, as well as a weighted variant of it, are derived. For details, see Step 8 in [BCI⁺20, Chapter 5] and more generally [BCI⁺20, Chapter 6] for regular correlated agreement, and [BCI⁺20, Section 7] for the weighted variant.

4.1 For linear subspaces ...

Let us now extend the list decoder analysis to an arbitrary linear subcode $\mathcal{C}'$ of $\mathcal{C} = \text{RS}_k[F, D]$. We directly do this for the general case $f_0, \dots, f_M \in F^D$ covered by Proposition 2. Assume that for every $z \in S$ as claimed we even have proximity to the subcode,

$$d(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, \mathcal{C}') < \theta,$$

and take any selection of proximates $P_z(X)$, $z \in S$, from the linear subspace $\mathcal{P}' \subseteq \mathcal{P}$ behind $\mathcal{C}'$. We claim that then, the polynomial

$$P(X, Z) = p_0(X) + Z \cdot p_1(X) + \dots + Z^M \cdot p_M(X)$$

from Proposition 2 *additionally satisfies that*

$$p_0(X), p_1(X), \dots, p_M(X) \in \mathcal{P}',$$

and thus belong to the subcode $\mathcal{C}'$.

To see this, assume that $\mathcal{C}'$ is generated by a single linear constraint, and thus its space of polynomials is $\mathcal{P}' = \{p \in \mathcal{P} : \Lambda p = 0\}$, where Λ is a linear

functional on $\mathcal{P} = F[X]^{<k}$. (The general case bears no additional difficulties.) Since

$$\frac{|S|}{2 \cdot \ell_m} \geq M \cdot \frac{(m + \frac{1}{2})^6}{6 \cdot \rho} \cdot |D|^2 \geq 3.7 \cdot M \cdot |D|^2 \quad (19)$$

even for the smallest choices of $|D|$ and m , regardless of $\rho < 1$, there exist at least $M + 1$ different points $z_0, z_1, \dots, z_M \in S$ for which $P(X, z_i) = P_{z_i}(X) \in \mathcal{P}'$. By linearity,

$$0 = \Lambda P(X, z_i) = \Lambda p_0(X) + z_i \cdot \Lambda p_1(X) + \dots + z_i^M \cdot \Lambda p_M(X),$$

for each z_i , and we conclude that

$$\Lambda p_0(X) = \Lambda p_1(X) = \dots = \Lambda p_M(X) = 0,$$

showing that all $p_i(X)$ belong to the subcode $\in \mathcal{C}'$, as claimed. (If $\mathcal{C}'$ is defined by several linear functionals, then $0 = \Lambda p_i(X)$ for each of the functionals Λ , yielding the same conclusion.)

From this sharpening of Proposition 2, both regular as well as weighted correlated agreement of $P(X, Z)$ are proven as in [BCI⁺20, Section 6 and 7], without any changes. We only cite the weighted variant; the regular case is already mentioned in Theorem 2. Given a sub-probability measure μ on D , and $f \in F^D$, we write

$$\text{agree}_\mu(f, \mathcal{C}') \geq 1 - \theta$$

if there exists a polynomial $p(X) \in \mathcal{P}'$ such that $\mu(\{x \in D : f(x) = p(x)\}) \geq 1 - \theta$.

Theorem 3. (*Weighted correlated agreement for subcodes*) Let $\mathcal{C}'$ be a linear subcode of $\text{RS}_k[F, D]$, and choose $\theta = 1 - \sqrt{\rho} \cdot (1 + \frac{1}{2 \cdot m})$, for some integer $m \geq 3$, where $\rho = k/|D|$. Assume a density function $\delta : D \rightarrow [0, 1] \cap \mathbb{Q}$ with common denominator $B \geq 1$, i.e. for all x in D

$$\delta(x) = \frac{m_x}{B},$$

for an integer value $m_x \in [0, B]$, and let μ be the sub-probability measure with density δ , defined by $\mu(\{x\}) = \delta(x)/|D|$. If for $f_0, f_1, \dots, f_M \in F^D$,

$$\frac{|\{z \in F : \text{agree}_\mu(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, \mathcal{C}') \geq 1 - \theta\}|}{|F|} > \varepsilon(\mathcal{C}, M, B, \theta),$$

where

$$\varepsilon(\mathcal{C}, M, B, \theta) = \frac{M}{|F|} \cdot \frac{(m + \frac{1}{2})}{\sqrt{\rho}} \cdot \max \left(\frac{(m + \frac{1}{2})^6}{3 \cdot \rho} \cdot |D|^2, \quad 2 \cdot (B \cdot |D| + 1) \right),$$

then there exist polynomials $p_0(X), p_1(X), \dots, p_M(X)$ belonging to the subcode $\mathcal{C}'$, and a set A with $\mu(A) \geq 1 - \theta$ on which $f_0, f_1, \dots, f_M$ coincide with $p_0(X), p_1(X), \dots, p_M(X)$, respectively.

4.2 ... and beyond

We emphasize the fact, that Proposition 2 applies also to a more general setting, in which the subspace of polynomials is characterized by a challenge-dependent function

$$\Gamma : \mathcal{P} \times F \longrightarrow F,$$

a polynomial (in the coordinates of an arbitrary basis) of typically small total degree $d = \deg \Gamma$, and

$$S = \left\{ z \in F : \exists P_z(X) \in \mathcal{P} \text{ s.t. } d(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, P_z) < \theta \wedge \Gamma(P_z, z) = 0 \right\},$$

where as before $\mathcal{P} = F[X]^{<k}$. In many applications, such as a Basefold proof for inner products with less-structured multilinear (see Section 6.1), we have $d = 2$.

As before, given the usual size bound (18) for S , and choosing the proximates accordingly (i.e., satisfying Γ), the polynomial $P(X, Z) = p_0(X) + Z \cdot p_1(X) + \dots + Z^M \cdot p_M(X)$ from Proposition 2 satisfies that

$$\Gamma(p_0 + z \cdot p_1 + \dots + z^M \cdot p_M, z) = 0$$

for z from a fraction of S , of size larger than

$$\frac{|S|}{2 \cdot \ell_m} \geq M \cdot \frac{(m + \frac{1}{2})^6}{6 \cdot \rho} \cdot |D|^2 > 3.7 \cdot M \cdot |D|^2$$

regardless of the choice of m and $\rho < 1$. Thus if $d \cdot M + 1 \leq 3.7 \cdot M \cdot |D|^2$ (which in the case $d = 2$ holds even for smallest domains), we obtain that

$$\Gamma(p_0 + Z \cdot p_1 + \dots + Z^M \cdot p_M, Z) = 0$$

as a formal identity. In other words, the θ -proximate $P(X, Z)$ satisfies the Γ -constraint over K , and in particular at every $Z = z$. Regular and weighted correlated agreement of $P(X, Z)$ is proven as before. We again cite only the weighted variant.

Theorem 4. (Non-linear generalization of Theorem 3) Let $\mathcal{C} = \text{RS}_k[F, D]$ be the Reed-Solomon code generated by the space of polynomials $\mathcal{P} = F[X]^{<k}$, and let $\Gamma : \mathcal{P} \times F \longrightarrow F$ be of total degree $d = 2$. As in Theorem 3, we take $\theta = 1 - \sqrt{\rho} \cdot (1 + \frac{1}{2 \cdot m})$ for some integer $m \geq 3$, where $\rho = k/|D|$, and let μ be the sub-probability measure with density $\delta : D \longrightarrow [0, 1] \cap \mathbb{Q}$. If for $f_0, f_1, \dots, f_M \in F^D$,

$$\left| \frac{\left\{ z \in F : \begin{array}{l} \exists P_z(X) \in \mathcal{P} \text{ s.t. } \Gamma(P_z, z) = 0 \\ \wedge \text{agree}_\mu(f_0 + z \cdot f_1 + \dots + z^M \cdot f_M, P_z) \geq 1 - \theta \end{array} \right\}}{|F|} \right| > \varepsilon(\mathcal{C}, M, B, \theta),$$

where

$$\varepsilon(\mathcal{C}, M, B, \theta) = \frac{M}{|F|} \cdot \frac{(m + \frac{1}{2})}{\sqrt{\rho}} \cdot \max \left(\frac{(m + \frac{1}{2})^6}{3 \cdot \rho} \cdot |D|^2, \quad 2 \cdot (B \cdot |D| + 1) \right),$$

with $B \geq 1$ being the common denominator of δ , then there exist polynomials $p_0(X), \dots, p_M(X) \in \mathcal{P}$ such that

$$\Gamma(p_0 + Z \cdot p_1 + \dots + Z^M \cdot p_M, Z) = 0 \in F[Z],$$

and which coincide with $f_0, f_1, \dots, f_M$, respectively, on a joint set A of weight $\mu(A) \geq 1 - \theta$.

Remark 5. The soundness error of the non-weighted variant of Theorem 4 is equal to $\varepsilon(\mathcal{C}, M, B, \theta)$ with $B = 1$.

5 Soundness in the oracle model

In this section we prove round-by-round soundness of the batched Basefold for Reed-Solomon codes, Protocol 2, as an interactive oracle proof of the target “language”

$$\mathcal{R} = \left\{ (g_0, \dots, g_M) : \begin{array}{l} \exists p_0, \dots, p_M \in \mathcal{P}_n \text{ s.t.} \\ d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta \\ \wedge \bigwedge_{k=0}^M P_k(\omega_1, \dots, \omega_M) = v_k, \\ \text{where } P_k = \Phi_n^{-1}(p_k) \end{array} \right\}, \quad (20)$$

for given words $g_0, \dots, g_M \in F^{D_0}$, query $\omega = (\omega_1, \dots, \omega_n) \in F^n$, and evaluation claims $v_0, \dots, v_M \in F$. In words, a tuple of words $(g_0, \dots, g_M)$ belongs to $\mathcal{R}$ if it has a θ -proximate $(p_0, \dots, p_M)$, in the correlated agreement sense, the multilinear representations $P_0, \dots, P_M$ of which satisfy the evaluation claims. The proximity parameter θ is taken from the list decoding regime, that is $\theta = 1 - \alpha$ with

$$\alpha = \left(1 + \frac{1}{2m} \right) \cdot \sqrt{\rho},$$

where $\rho = 2^n/|D|$ is the rate of the Reed-Solomon code $\mathcal{C} = \text{RS}_{2^n}[F, D]$, and $m \geq 3$ is the multiplicity parameter of the Guruswami-Sudan list decoder. Although we confine ourselves to the list decoding regime, we stress the fact that this is merely for brevity. The entire analysis can be carried over verbatim to the unique decoding regime, using the respective adaptations of Theorem 2 and Theorem 3.

As outlined in Section 2, we consider an interactive oracle proof as a chain of randomized reductions, in our case

$$\mathcal{R} \xrightarrow{\lambda_0} \mathcal{R}_0 \xrightarrow{\lambda_1} \mathcal{R}_1 \xrightarrow{\lambda_2} \dots \xrightarrow{\lambda_n} \mathcal{R}_n \xrightarrow{Q} \mathcal{R}_V, \quad (21)$$

which gradually reduce the target claim $(g_0, \dots, g_M) \in \mathcal{R}$ to the smaller claims that the prover messages $(f_0, A_0, \dots, f_i, A_i)$ belong to

$$\mathcal{R}_i = \mathcal{R}_i(\lambda_0, \dots, \lambda_i),$$

to be defined below, and where

$$\mathcal{R}_V = \mathcal{R}_V(\lambda_0, \dots, \lambda_n, Q)$$

is the final verifier relation, consisting of the verifier consistency checks, using the queries $x \in Q$ sampled in the query phase. (In the last round $i = n$, there is no refinement polynomial A_i .) Besides the consistency of the sumcheck polynomials provided so far,

$$q_{j-1}(\lambda_j) = q_j(0) + q_j(1), \quad j = 0, \dots, i-1, \quad (22)$$

each $\mathcal{R}_i$ requires the current $f_i \in F^{D_i}$ having a θ -proximate $p_i(X) \in \mathcal{P}_{n-i}$ the multilinear representation of which, $P_i = \Phi_i^{-1}(p_i)$, satisfies the current specialization of the sumcheck,

$$\begin{aligned} L((\omega_1, \dots, \omega_i), (\lambda_1, \dots, \lambda_i)) \cdot P_i(\omega_{i+1}, \dots, \omega_n) &= q_{i-1}(\lambda_i) \\ &= L((\omega_1, \dots, \omega_i), (\lambda_1, \dots, \lambda_i)) \cdot A_{i-1}(\lambda_i), \end{aligned} \quad (23)$$

and $(g_0, \dots, g_M, f_0, \dots, f_i)$ are sufficiently consistent under folding by $\lambda_0, \dots, \lambda_i$. Concretely, we require that

$$\left| \left\{ x \in D_0 : \begin{array}{l} \text{With } x_j = \pi^j(x), j = 0, \dots, i, \\ f_0(x_0) = \sum_{k=0}^M \lambda_0^k \cdot g_k(x_0) \wedge \\ \bigwedge_{j=1}^i f_j(x_j) = \text{fold}_{\lambda_j}(f_{j-1})(x_j) \\ \wedge \quad f_i(x_i) = p_i(x_i) \end{array} \right\} \right| \geq \alpha \cdot |D_0|, \quad (24)$$

with $\alpha = 1 - \theta$, where $\text{fold}_{\lambda_j}(f_{j-1})$ is the folding of f_{j-1} using randomness λ_j . In full extent, the set $\mathcal{R}_i = \mathcal{R}_i(\lambda_0, \dots, \lambda_i)$ is defined as

$$\mathcal{R}_i = \left\{ \begin{array}{l} (g_0, \dots, g_M) \\ (f_0, A_0, \dots, f_i, A_i) \end{array} : \begin{array}{l} \exists p_i(X) \in \mathcal{P}_{n-i}, P_i = \Phi_{n-i}^{-1}(p_i), \\ \text{so that (22), (23), (24) holds with} \\ \alpha = 1 - \theta \end{array} \right\}, \quad (25)$$

for $0 \leq i \leq n-1$. In the last step $i = n$, there is no refinement polynomial λ_n , and the sumcheck equation (23) collapses to

$$\begin{aligned} L((\omega_1, \dots, \omega_n), (\lambda_1, \dots, \lambda_n)) \cdot P_n &= q_{n-1}(\lambda_n) \\ &= L((\omega_1, \dots, \omega_n), (\lambda_1, \dots, \lambda_n)) \cdot A_{n-1}(\lambda_n), \end{aligned} \quad (26)$$

with P_n being the constant $\Phi_0^{-1}(p_n)$, and we moreover demand that $f_n = p_n$ everywhere on D_n .

$$\mathcal{R}_n = \left\{ \begin{array}{l} (g_0, \dots, g_M) \\ (f_0, A_0, \dots, f_n) \end{array} : \begin{array}{l} f_n = c \in \mathcal{P}_{n-i}, P_n = \Phi_0^{-1}(c), \\ \text{so that (26), and (22), (24) holds for} \\ i = n, \alpha = 1 - \theta \end{array} \right\}. \quad (27)$$

In the following three lemmas, Lemma 1, Lemma 2 and Lemma 3, we quantify the soundness errors of each of the reduction steps in (21), altogether proving round-by-round soundness of the protocol.

5.1 Batching step

As sketched in Section 3.2, the correlated agreement theorem with respect to the linear subspaces

$$\mathcal{C}'_i < \mathcal{C}_i = \text{RS}[F, D_i, 2^{n-k}],$$

generated by the constrained set of polynomials

$$\mathcal{P}'_{n-i} = \{p(X) \in \mathcal{P}_{n-i} : P(\omega_{i+1}, \dots, \omega_n) = 0\}, \quad (28)$$

will play a crucial role in the soundness analysis. This is the “punctured” space of all polynomials from $\mathcal{P}_{n-i}$, the multilinear representation of which has a zero at the given query.

Lemma 1 (Soundness batching step). *Round $i = 0$ in Protocol 2 as a randomized reduction $\mathcal{R} \xrightarrow{\lambda_0} \mathcal{R}_0$ has soundness error*

$$\varepsilon_0 = \varepsilon(\mathcal{C}_0, M, \theta),$$

where $\varepsilon(\mathcal{C}_0, M, \theta)$ is the soundness error from Theorem 2.

Proof. Assume that with probability greater than $\varepsilon_0 = \varepsilon(\mathcal{C}_0, M, \theta)$, there exists a potential prover answer (f_0, Λ_0) extending $(g_0, \dots, g_M)$ to an element from $\mathcal{R}_0$ as defined above. Then the linear combination of the centered functions $g'_k = g_k - v_k \cdot \Phi_n(1)$ are θ -proximate to the subspace $\mathcal{C}'_0$ of $\mathcal{C}_0$, with probability

$$\Pr_{\lambda_0} \left[\exists p'_0 \in \mathcal{P}'_n \text{ s.t. } \text{agree} \left(\sum_{k=0}^M g'_k \cdot \lambda_0^k, p'_0(X) \right) \geq \alpha \right] > \varepsilon(\mathcal{C}_0, M, \theta).$$

By the correlated agreement theorem for subspaces, Theorem 2, we conclude that there exist polynomials

$$p'_0(X), \dots, p'_M(X) \in \mathcal{P}'_n$$

which agree with $g_0, \dots, g_M$ on a joint set of density $\geq \alpha$. Over the same set the non-centered polynomials

$$p'_0(X) + v_0 \cdot \Phi_n(1), \dots, p'_M(X) + v_M \cdot \Phi_n(1) \in \mathcal{P}_n$$

agree with $g_0, \dots, g_M$. Their multilinear representations $P_k \in F[X_1, \dots, X_n]$ satisfy $P_k(\omega) = v_k$, showing that $(g_1, \dots, g_M) \in \mathcal{R}$. $\square$

5.2 Folding steps

For the analysis of the folding rounds it is convenient to paraphrase the folding consistencies (24) via a *subprobability measure* μ_i on D_i ,

$$\mu_i(\{y\}) = \frac{\delta_i(y)}{|D_i|},$$

with $\delta_i(y)$ being the expected consistency “above” the point $y \in D_i$. These conditional probabilities can be recursively defined as follows. Over D_0 ,

$$\delta_0(x) := \begin{cases} 1 & \text{if } f_0(x) = \sum_{k=0}^M \lambda_0^k \cdot g_k(x) \\ 0 & \text{otherwise,} \end{cases} \quad (29)$$

and for $i = 1, \dots, n$, the probabilities over D_i are formed from the probabilities over D_{i-1} , in conjunction with the consistency of the most recent folding,

$$\delta_i(x) := \begin{cases} \delta_{i-1}^*(x) & \text{if } f_i(x) = (1 - \lambda_i) \cdot f_{i-1,0}(x) + \lambda_i \cdot f_{i-1,1}(x) \\ 0 & \text{otherwise,} \end{cases} \quad (30)$$

where $\delta_{i-1}^*(x^2) = \frac{1}{2} \cdot (\delta_{i-1}(x) + \delta_{i-1}(-x))$. With the consistency measure μ_i , we rewrite the i -th “language” as

$$\mathcal{R}_i = \left\{ \begin{array}{l} (g_0, \dots, g_M) \\ (f_0, A_0, \dots, f_i, A_i) \end{array} : \begin{array}{l} \exists p_i(X) \in \mathcal{P}_{n-i}, P_i = \Phi_{n-i}^{-1}(p_i), \\ \text{agree}_{\mu_i}(f_i, p_i) \geq 1 - \theta, \\ \text{and (22), (23) holds} \end{array} \right\}, \quad (31)$$

for $i = 0, \dots, n-1$, and like wise for the last relation $\mathcal{R}_n$.

Lemma 2 (Soundness folding steps). *Round $i \in \{1, \dots, n\}$ in Protocol 2 as a randomized reduction $\mathcal{R}_{i-1} \xrightarrow{\lambda_i} \mathcal{R}_i$ has soundness error*

$$\varepsilon_i = \frac{1}{|F|} + \varepsilon(\mathcal{C}_i, 1, B_i, \theta),$$

where $\varepsilon(\mathcal{C}_0, 1, B_i, \theta)$ is the soundness error from Theorem 3, with $B_i = |D|/|D_i| = 2^i$.

Proof. Assume that with probability greater than ε_i as claimed, and in the verifier challenge λ_i , there exists a potential prover answer (f_i, A_i) which extends the previous messages to an element of $\mathcal{R}_i$, as defined in (31). For each of these λ_i , there exists a sumcheck compliant polynomial $p_i(X) \in \mathcal{P}_{n-i}$ with $\text{agree}_{\mu_i}(f_i, p_i) \geq 1 - \theta$. Restricting $A = \{x \in D_i : f_i(x) = p_i(x)\}$ to the subset A' where $\delta_i > 0$, and therefore the last folding check holds, we see that

$$\mu_{i-1}(\pi^{-1}(A')) = \mu_i(A') = \mu_i(A),$$

and therefore

$$\text{agree}_{\mu_{i-1}^*}((1 - \lambda_i) \cdot f_{i-1,0} + \lambda_i \cdot f_{i-1,1}, p_i) \geq 1 - \theta,$$

where μ_{i-1}^* is the projection of μ_{i-1} onto D_i , defined by $\mu_{i-1}^*(A_i) = \mu_{i-1}(\pi^{-1}(A_i))$.

Using sumcheck compliance as in our proof sketch from Section 3.2, we conclude that except with probability $1/|F|$ in λ_i (for the possible zero of the Lagrangian), hence still of probability greater than

$$\varepsilon_i - \frac{1}{|F|} = \varepsilon(\mathcal{C}_i, 1, B_i, \theta),$$

the centered polynomial $p'_i = p_i - \Lambda_{i-1}(\lambda_i) \cdot \Phi_{n-i}(1)$ is from the subspace $\mathcal{P}'_{n-i}$ as defined above, and the centered parts

$$\begin{aligned} f'_{i-1,0} &= f_{i-1,0} - \Lambda_{i-1}(0) \cdot \Phi_{n-i}(1), \\ f'_{i-1,1} &= f_{i-1,1} - \Lambda_{i-1}(1) \cdot \Phi_{n-i}(1), \end{aligned}$$

satisfy

$$\text{agree}_{\mu_{i-1}^*}((1 - \lambda_i) \cdot f'_{i-1,0} + \lambda_i \cdot f'_{i-1,1}, p'_i) \geq 1 - \theta.$$

Using the weighted correlated agreement theorem for subspaces, Theorem 3, we conclude that $f'_{i-1,0}$ and $f'_{i-1,1}$ agree with some

$$p'_{i-1,0}(X), p'_{i-1,1}(X) \in \mathcal{P}'_{n-i}$$

on a set $A_i \subseteq D_i$ of weight $\mu_{i-1}^*(A_i) \geq 1 - \theta$. Over the same set, the non-centered parts $f_{i-1,0}$ and $f_{i-1,1}$ agree with

$$\begin{aligned} p_{i-1,0}(X) &= p'_{i-1,0}(X) + \Lambda_{i-1}(0) \cdot \Phi_{n-i}(1), \\ p_{i-1,1}(X) &= p'_{i-1,1}(X) + \Lambda_{i-1}(1) \cdot \Phi_{n-i}(1), \end{aligned}$$

both polynomials from $\mathcal{P}_{n-i}$, the multilinear representations $P_{i-1,0}, P_{i-1,1}$ of which satisfy

$$\begin{aligned} P_{i-1,0}(\omega_{i+1}, \dots, \omega_n) &= \Lambda_{i-1}(0), \\ P_{i-1,1}(\omega_{i+1}, \dots, \omega_n) &= \Lambda_{i-1}(1). \end{aligned}$$

(In the edge case $i = n$ the two latter equations collapse to $P_{n-1,0} = \Lambda_{n-1}(0)$ and $P_{n-1,1} = \Lambda_{n-1}(1)$ as constants.) Over the preimage $A_{i-1} = \pi^{-1}(A_i)$ the word f_{i-1} coincides with the values of

$$p_{i-1}(X) = p_{i-1,0}(X^2) + X \cdot p_{i-1,1}(X^2) \in \mathcal{P}_{n-i+1},$$

and by construction its multilinear representation P_{i-1} satisfies

$$\begin{aligned} P_{i-1}(\omega_i, \omega_{i+1}, \dots, \omega_n) &= (1 - \omega_i) \cdot \Lambda_{i-1}(0) + \omega_i \cdot \Lambda_{i-1}(1) \\ &= L(\omega_i, 0) \cdot \Lambda_{i-1}(0) + L(\omega_i, 1) \cdot \Lambda_{i-1}(1), \end{aligned}$$

and hence is sumcheck compliant,

$$\begin{aligned} L(\omega_1, \dots, \omega_{i-1}, \lambda_1, \dots, \lambda_{i-1}) \cdot P_{i-1}(\omega_i, \omega_{i+1}, \dots, \omega_n) \\ = q_{i-1}(0) + q_{i-1}(1) = q_{i-2}(\lambda_{i-1}). \end{aligned}$$

Since $\text{agree}_{\mu_{i-1}^*}(f_{i-1}, p_{i-1}) = \mu_{i-1}^*(A_i) \geq 1 - \theta$, the prover message $(f_0, \Lambda_0, \dots, f_{i-1}, \Lambda_{i-1})$ belongs to $\mathcal{R}_{i-1}$. $\square$

5.3 Query phase

The final step in Protocol 2 is the query phase, a probabilistic reduction in the random queries from D , which reduces the claim that $(f_0, A_0, \dots, f_n) \in \mathcal{R}_n$ to that it belongs to the verifier relation $\mathcal{R}_V = \mathcal{R}_V(\lambda_0, \dots, \lambda_n, Q)$, which involve only the sumcheck consistencies (22), (26) and the verifier spot-checks in the definition of (24), for each x from the set of drawn queries Q . In full extent,

$$\mathcal{R}_V = \left\{ \begin{array}{l} (g_0, \dots, g_M) : \begin{array}{l} f_n = c \in \mathcal{P}_0, P_n = \Phi_0^{-1}(c), \text{ s.t. (26), (22) for } i = n, \\ \text{and for all } x \in Q, \text{ with } x_j = \pi^j(x), \\ f_0(x_0) = \sum_{k=0}^M \lambda_0^k \cdot g_k(x_0) \text{ and} \\ f_j(x_j) = \text{fold}_{\lambda_j}(f_{j-1})(x_j) \text{ for } j = 0, \dots, n \end{array} \end{array} \right\}.$$

Lemma 3 (Soundness query phase). *The query phase in Protocol 2 as a randomized reduction $\mathcal{R}_n \xrightarrow{x_1, \dots, x_s} \mathcal{R}_V$ has soundness error $\varepsilon_Q = (1 - \theta)^s$.*

Proof. The soundness error corresponds to the probability that a collected prover message $(f_0, A_0, \dots, f_n) \notin \mathcal{R}_n(\lambda_0, \dots, \lambda_n)$ is not detected by s random samples. We have $(f_0, A_0, \dots, f_n) \notin \mathcal{R}_n$, if either $f_n \notin \mathcal{C}_n$, or one of the sumcheck consistency checks fail, or the folding consistency set defined in (24), with $i = n$, is of density $< 1 - \theta$. The former two conditions are always detected, and the latter is detected except with probability $\varepsilon_Q \leq (1 - \theta)^s$. $\square$

5.4 Skipping commitments

A common strategy to reduce the proof size of FRI-like protocols, is to do higher reduction steps in the folding cascade. In Basefold, this corresponds to running several rounds of the sumcheck, say $r \geq 2$, without committing to the intermediate foldings. In the query phase, the verifier then uses the last commitment f_i to check consistency with the next commitment f_{i+r} via r -fold folding identity

$$\begin{aligned} f_{i+r} &= \text{fold}_{\lambda_{i+1}, \dots, \lambda_{i+r}}(f_i) \\ &= \text{fold}_{\lambda_{i+r}} \circ \text{fold}_{\lambda_{i+r-1}} \circ \dots \circ \text{fold}_{\lambda_{i+1}}(f_i). \end{aligned} \tag{32}$$

The soundness analysis can be carried over without any changes, thinking of the non-committed foldings $f_{i+1}, \dots, f_{i+r-1}$ as virtual commitments derived from f_i , still keeping them in the definition of the intermediate sets $\mathcal{R}_{i+1}, \dots, \mathcal{R}_{i+r-1}$.

6 Generalizations

In this section we discuss several generalizations of Basefold, and how the soundness analysis from Section 5 extends to these cases. First of all, in Section 6.2, we quickly discuss the adaption of Basefold to different FFT environments, with emphasis on the additive FFT over binary fields. Then in Section 6.1, we generalize the type of sumcheck expressions to be proven by Basefold, covering multi-query evaluation proofs as well as the recent FRI-Binius optimization [Dia24b], which we shortly discuss in Section 6.3.

6.1 More expressive inner products

Protocol 1 and 2 are easily extended to more expressive hypercube sums than the evaluation inner product for a single query on the multilinear $G_0, \dots, G_M$. For simplicity, we restrict to assertions of the form

$$\langle G_k, R \rangle_{H_n} = v_k, \quad (33)$$

sharing the same $R \in F[X_1, \dots, X_n]$ for $k = 0, \dots, M$, any succinct evaluable multilinear. This case covers linear combinations of Lagrangians, as used in multi-point evaluation proofs, or WHIR, as well as the row-batched optimization of FRI-Binius, which we address in Section 6.4 and 6.3.

For assertions of the form (33), the sumcheck is still quadratic, but since R is in general not decomposable as a Lagrangian $L(\cdot, \omega)$, we cannot work with linear refinement polynomials as we did in Section 3. This rules out a reduction of the soundness analysis to correlated agreement for linear subcodes, Theorem 3. Instead, we return to the regular sumcheck convention and let the prover directly provide the quadratic polynomials

$$q_i(X) = \langle P(\lambda_1, \dots, \lambda_i, X, \cdot), R(\lambda_1, \dots, \lambda_i, X, \cdot) \rangle_{H_{n-i-1}},$$

in each of the rounds $i = 0, \dots, n-1$, and correlated agreement with the needed additional algebraic properties will be guaranteed by the non-linear generalization Theorem 4.

The differences to Section 5 are merely technical. The “language” to be proven is

$$\mathcal{R} = \left\{ (g_0, \dots, g_M) : \begin{array}{l} \exists p_0, \dots, p_M \in \mathcal{P}_n \text{ s.t.} \\ d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta \\ \wedge \bigwedge_{k=0}^M \langle P_k, R \rangle_{H_n} = v_k \end{array} \right\}, \quad (34)$$

and the further “languages”

$$\mathcal{R}_i = \left\{ \begin{array}{l} (g_0, \dots, g_M) \\ (f_0, q_0, \dots, f_i, q_i) \end{array} : \begin{array}{l} \exists p_i(X) \in \mathcal{P}_{n-i}, P_i = \Phi_{n-i}^{-1}(p_i), \\ \text{agree}_{\mu_i}(f_i, p_i) \geq 1 - \theta, \\ \text{and (22), (23) holds} \end{array} \right\}, \quad (35)$$

for $i = 0, \dots, n-1$, where we address only the first line in (23), and likewise for the last “language” $\mathcal{R}_n$.

Soundness of the batching step $\mathcal{R} \xrightarrow{\lambda_0} \mathcal{R}_1$ is proven as in Lemma 1, using the linear subspace

$$\mathcal{P}'_n = \{p(X) \in \mathcal{P}_n : \langle \Phi_n^{-1}(p), R \rangle_{H_n} = 0\},$$

yielding the same soundness error. For the folding steps we obtain slightly smaller errors compared to Lemma 2, which attributes the fact that Theorem 4 directly handles the quadratic refinement polynomials.

Lemma 4 (Soundness folding steps). *Round $i \in \{1, \dots, n\}$ in Protocol 2 as a randomized reduction $\mathcal{R}_{i-1} \xrightarrow{\lambda_i} \mathcal{R}_i$ has soundness error*

$$\varepsilon_i = \varepsilon(\mathcal{C}_i, 1, B_i, \theta),$$

where $\varepsilon(\mathcal{C}_0, 1, B_i, \theta)$ is the soundness error from Theorem 4, with $B_i = |D|/|D_i| = 2^i$.

Proof. Assume that with probability greater than ε_i as claimed, and in the verifier challenge λ_i , there exists a potential prover answer (f_i, q_i) which extends the previous messages $(g_0, \dots, g_M, f_0, q_0, \dots, f_{i-1}, q_{i-1})$ to an element of $\mathcal{R}_i$, as defined in (35). (In the last step $i = n$ there is no q_i .) As the proof of Lemma 4 we argue that for any such λ_i , there exists a sumcheck compliant polynomial $p_i(X) \in \mathcal{P}_{n-i}$ so that

$$\text{agree}_{\mu_{i-1}^*}((1 - \lambda_i) \cdot f_{i-1,0} + \lambda_i \cdot f_{i-1,1}, p_i) \geq 1 - \theta.$$

Sumcheck-compliance means that

$$\Gamma_i(p_i, \lambda_i) = 0,$$

where $\Gamma_i : \mathcal{P}_{n-i} \times F \rightarrow F$ is the non-linear constraint

$$\Gamma_i(p_i, z) = \langle P_i, R(\lambda_1, \dots, \lambda_{i-1}, z, \cdot) \rangle_{H_{n-i}} - q_{i-1}(z)$$

with $P_i = \Phi_{n-i}^{-1}(p_i)$, and $q_{i-1}(z)$ the claimed sumcheck polynomial. (In the edge case $i = n$, $\mathcal{P}_{n-i}$ is the space of constants, and the inner product reduces to the singleton sum $p_n \cdot R(\lambda_1, \dots, \lambda_{n-1}, z)$.) The total degree of Γ_i is $d = 2$, and we apply Theorem 4 to conclude that $f_{i-1,0}$ and $f_{i-1,1}$ agree over a joint set A_i of density $\mu_i(A_i) \geq 1 - \theta$ with polynomials

$$p_{i-1,0}(X), p_{i-1,1}(X) \in \mathcal{P}_{n-i},$$

satisfying $\Gamma_i((1 - Z) \cdot p_{i-1,0} + Z \cdot p_{i-1,1}, Z)$ as formal identity. In particular $\Gamma_i(p_{i-1,0}, 0) = \Gamma_i(p_{i-1,1}, 1) = 0$, meaning that their multilinear representations $P_{i-1,0}$ and $P_{i-1,1}$ satisfy

$$\begin{aligned} \langle P_{i-1,0}, R(\lambda_1, \dots, \lambda_{i-1}, 0) \rangle_{H_{n-i}} &= q_{i-1}(0), \\ \langle P_{i-1,1}, R(\lambda_1, \dots, \lambda_{i-1}, 1) \rangle_{H_{n-i}} &= q_{i-1}(1). \end{aligned}$$

The rest of the proof is again identical to that of Lemma 4. Over the preimage $A_{i-1} = \pi^{-1}(A_i)$, a set of measure $\mu_{i-1}(A_i) \geq 1 - \theta$, the word f_r coincides with the values of

$$p_{i-1}(X) = p_{i-1,0}(X^2) + X \cdot p_{i-1,1}(X^2) \in \mathcal{P}_{n-i+1},$$

and by construction its multilinear representation P_{i-1} satisfies

$$\begin{aligned} \langle P_{i-1}, R(\lambda_1, \dots, \lambda_{i-1}, \cdot) \rangle_{H_{n-i+1}} &= \langle P_{i,0}, R(\lambda_1, \dots, \lambda_{i-1}, 0) \rangle_{H_{n-i}} \\ &+ \langle P_{i,1}, R(\lambda_1, \dots, \lambda_{i-1}, 1) \rangle_{H_{n-i}} = q_{i-1}(0) + q_{i-1}(1) = q_{i-2}(\lambda_{i-1}), \end{aligned}$$

and hence is sumcheck compliant. Altogether, $(g_0, \dots, g_M, f_0, q_0, \dots, f_{i-1}, q_{i-1})$ belongs to $\mathcal{R}_{i-1}$. $\square$

6.2 Other FFTs

Adapting Basefold to other FFT-encodable algebraic geometry codes is merely a technical step. For simplicity, and in particular in view of Section 6.3, we restrict ourselves to Reed-Solomon codes over binary fields, with the *additive FFT* [LCH14] used for encoding. Other codes such as EC-FFT codes [BCKL21] or Circle Codes [HLP24] are treated likewise.

Let F be a finite binary field, i.e. a finite field of characteristic 2. Instead of subgroups of the multiplicative group, the additive FFT takes subgroups of the additive group, i.e. $\mathbb{F}_2$ -linear subspaces of F as the domains. Similar to the regular case, the FFT works along a chain of projected subspaces

$$U = U_0 \xrightarrow{\pi_1} U_1 \xrightarrow{\pi_2} \dots \xrightarrow{\pi_n} U_n, \quad (36)$$

where

$$\pi_i(x) = x \cdot (x - b_i),$$

with $b_i \in F$, are suitably chosen quadratic maps⁷ which halve the size of the subspaces in each of the steps, until ending up with a singleton domain U_n . (Recall that quadratic maps are $\mathbb{F}_2$ -linear.) As “twiddle functions” $t_i : U_i \rightarrow F$ one may take again $t_i(X) = X$, for $i = 0, \dots, n-1$, or a properly normalized⁸ variant if one desires an optimized butterfly network.

Given a function $f \in F^U$ over $U = U_0$, with values in F (or more generally, an extension of F), the additive FFT computes the coefficients of the interpolant

$$p(X) = \sum_{i=0}^{2^n-1} c_i \cdot b_{n,i}(X)$$

with respect to a polynomial basis $\mathcal{B}_n = \{b_{n,i}(X)\}$, which is different to the monomial one, defined by the projection chain and the twiddle functions,

$$b_{n,i}(X) = t_0(X)^{i_0} \cdot (t_1 \circ \pi_1)(X)^{i_1} \cdot \dots \cdot (t_{n-1} \circ \pi_{n-1} \circ \dots \circ \pi_1)(X)^{i_{n-1}},$$

where $(i_0, \dots, i_{n-1})$ are the bits of $i = \sum_{k=0}^{n-1} i_k \cdot 2^k$. By the degrees of t_i and π_i , these polynomials are of degree less than 2^n and hence form a basis of $\mathcal{P}_n$. As in the multiplicative case, the algorithm is based on the decomposition of a function $f_i \in K^{U_i}$ into “even” and “odd” parts,

$$f_i(x) = f_{i,0}(\pi_{i+1}(x)) + t_i(x) \cdot f_{i,1}(\pi_{i+1}(x)), \quad (37)$$

which are computed by value, using a “butterfly” along the fibers of the projection, which are of the form $\{x, x + b_{i+1}\}$, see formula (38) below.

⁷ In fact, if $U_i = u_i + V_i$ with V_i being a non-affine subspace, then any quadratic map of the above form, which annihilates a basis vector of V_i , is fine.

⁸ The proper normalization is so that $t_i(x + b_{i+1}) = t_i(x) + 1$. Equivalently, one may integrate normalization into the projection, see [GM10, LCH14, DP24].

Likewise *additive FRI* [BBHR18], which proves proximity to the Reed-Solomon code $\mathcal{C}_0 = \text{RS}_{2^n}[F, D]$ over an evaluation domain D , a disjoint coset union of U_0 , goes along the projected domains

$$D = D_0 \xrightarrow{\pi_1} D_1 \xrightarrow{\pi_2} \dots \xrightarrow{\pi_n} D_n,$$

which are also halved in each step. Starting with $f_0 \in F^D$, FRI recursively takes random linear combinations of the “even” and “odd” parts,

$$f_{i+1}(\pi_{i+1}(x)) = f_{i,0}(\pi_{i+1}(x)) + \lambda_i \cdot f_{i,1}(\pi_{i+1}(x)),$$

thereby reducing the initial proximity claim to gradually simpler claims, which are with respect to the codes $\mathcal{C}_i = \text{RS}_{2^{n-i}}[F, D_i]$ over the projected domains, for $i = 1, \dots, n$.

In regards to Basefold, a multilinear $P \in F[X_1, \dots, X_n]$ is again represented as a univariate polynomial $p(X) \in \mathcal{P}_n$ via

$$p(X) = \sum_{i=0}^{2^n-1} P(i_0, \dots, i_{n-1}) \cdot b_{n,i}(X),$$

and committed as word from $\mathcal{C}_0$. By the recursive structure of the polynomial basis, this identification preserves the connection between partial substitution and FRI-like folding of even and odd parts. With this different FFT environment, Protocol 1 and 2 remain essentially unchanged, except that “even” and “odd” parts are computed (and verified) via

$$\begin{pmatrix} f_{i,0}(\pi_{i+1}(x)) \\ f_{i,1}(\pi_{i+1}(x)) \end{pmatrix} = \begin{pmatrix} 1 & t_i(x) \\ 1 & t_i(x + b_{i+1}) \end{pmatrix}^{-1} \cdot \begin{pmatrix} f_i(x) \\ f_i(x + b_{i+1}) \end{pmatrix}. \quad (38)$$

The subcode correlated agreement theorem from Section 4.1 is valid for fields of arbitrary characteristic (as does its non-linear generalization), and hence Theorem 1 with its proof from Section 5 holds verbatim.

6.3 FRI-Binius

FRI-Binius [DP24] is a *packed polynomial commitment scheme*, in which multilinear over some small field $\mathbb{F}_q$ (for example $\mathbb{F}_q = \mathbb{F}_2$) are committed via a “packed” representation over some larger field

$$E > \mathbb{F}_q.$$

We restrict ourselves to the case of binary fields and Reed-Solomon codes, and for brevity we omit certain details which are not essential for the core of the construction.

Let $\mathbb{F}_q$ be a field of characteristic two, and E be an extension field of dimension $d = \dim E/\mathbb{F}_q$, with an (additive) FFT domain D of size $|D| > n$. Fix a

basis $\{\beta_1, \dots, \beta_d\}$ of $E/\mathbb{F}_q$, then any collection⁹ of small-field multilinear

$$P_1, \dots, P_d \in \mathbb{F}_q[X_1, \dots, X_n]$$

is committed via the “packed” multilinear

$$P = \sum_{i=1}^d \beta_i \cdot P_i \in E[X_1, \dots, X_n], \quad (39)$$

over the larger field E , as code word f from $\mathcal{C} = \text{RS}_{2^n}[E, D]$ the Reed-Solomon code of rate $\rho = 2^n/|D|$, using the same identification of multilinear and univariates as before. Conversely, given any $P \in E[X_1, \dots, X_n]$ it can be uniquely decomposed into component polynomials $P_1, \dots, P_d \in \mathbb{F}_q[X_1, \dots, X_n]$ satisfying (39).

Now, for a given a query $\omega \in F^n$ from any other field $F \geq \mathbb{F}_q$ (the *query field*, a cryptographically large extension), we wish to prove the values

$$v_i = P_i(\omega) \in F, \quad i = 1, \dots, d, \quad (40)$$

by means of the packed polynomial P . While in [DP24], these claims are expressed as an evaluation inner product over the tensor algebra $F \otimes E$, we largely avoid the notion of tensor algebras, and directly describe the row-batched optimization from [Dia24b]. Let $d' = \dim F/\mathbb{F}_q$ and $\{\gamma_1, \dots, \gamma_{d'}\}$ be a basis of $F/\mathbb{F}_q$, and write

$$v_i = \sum_{j=1}^{d'} v_{i,j} \cdot \gamma_j \quad \text{and} \quad L(\omega, \cdot) = \sum_{j=1}^{d'} \gamma_j \cdot L_{\omega,j}(\cdot), \quad (41)$$

with $v_{i,j} \in \mathbb{F}_q$ and component multilinear $L_{\omega,j} \in \mathbb{F}_q[X_1, \dots, X_n]$. Then the claims (40) are equivalent to that

$$\langle P, L_{\omega,j} \rangle_{H_n} = \sum_{i=1}^d \beta_i \cdot \langle P_i, L_{\omega,j} \rangle_{H_n} = \sum_{i=1}^d \beta_i \cdot v_{i,j},$$

for every $j = 1, \dots, d'$, which is then proven via the random linear combination

$$\left\langle P, \sum_{j=1}^{d'} \lambda^{j-1} \cdot L_{\omega,j} \right\rangle_{H_n} = \sum_{j=1}^{d'} \lambda^{j-1} \cdot \sum_{i=1}^d \beta_i \cdot v_{i,j}, \quad (42)$$

with $\lambda \leftarrow \$ F$. This is done using Basefold from Section 6.1, for fields of characteristic 2 adapted to the additive FFT as described in Section 6.2. Note that the polynomials $L_{\omega,j}$ are multilinear, and more importantly, they can be evaluated simultaneously, *still in a succinct manner*, at any point $\lambda \in F^n$. (This can be done by means of tensor algebra operations¹⁰, see [DP24].) The protocol for a batch of packed multilinear is as follows.

⁹ In FRI-Binius these polynomials are a Lagrange decomposition of a larger multilinear over F .

¹⁰ The values $L_{\omega,j}(\lambda)$ are the rows of the Lagrangian $L(\omega^t, \lambda)$ evaluated in the tensor algebra $F \otimes K$, where ω^t is an element in the vertical embedding of F , and λ is an element in the horizontal embedding of K .

Protocol 3 (Row-batched FRI-Binius [Dia24b]). *Let $\mathbb{F}_q \leq E \leq F$ as above. Given multilinear $G_0, \dots, G_M \in E[X_1, \dots, X_n]$ over E , committed as Reed-Solomon code words $g_0, \dots, g_M \in \mathcal{C} = \text{RS}_{2^n}[E, D]$, and evaluation claims $v_i^{(k)} \in F$, $i = 1, \dots, d$, $k = 0, \dots, M$ for their component polynomials $G_{k,i}$, at some query $\omega \in F^n$.*

1. *The verifier sends a random $\lambda \leftarrow \$ F$ to the prover.*
2. *Both prover and the verifier now run Basefold from Section 6.1, with proximity parameter $\theta = (1 + \frac{1}{2m}) \cdot \sqrt{\rho}$ and $s \geq 1$ samples, on $G_0, \dots, G_M$ for their inner products $\langle G_k, L_{\omega, \lambda} \rangle_{H_n} = v_{\lambda}^{(k)}$, $0 \leq k \leq M$, where*

$$L_{\omega, \lambda} = \sum_{j=1}^{d'} \lambda^{j-1} \cdot L_{\omega, j}, \quad v_{k, \lambda} = \sum_{j=1}^{d'} \sum_{i=1}^d \lambda^{j-1} \cdot \beta_i \cdot v_{i, j}^{(k)}, \quad (43)$$

using the decomposition of L_{ω} and $v_i^{(k)}$ as in (41).

Protocol 3 is an interactive oracle proof for the “language”

$$\mathcal{R} = \left\{ (g_0, \dots, g_M) : \begin{array}{l} \exists p_0, \dots, p_M \in E[X]^{<2^n} \text{ s.t.} \\ d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta \\ \wedge \bigwedge_{k=0}^M \bigwedge_{i=1}^d P_{k,i}^{[E:\mathbb{F}_q]}(\omega_1, \dots, \omega_n) = v_i^{(k)} \end{array} \right\}, \quad (44)$$

where the $P_{k,i}$ are the component polynomials of P_k the multivariate representations of p_k . The first round reduces $\mathcal{R}$ to

$$\mathcal{R}' = \left\{ (g_0, \dots, g_M) : \begin{array}{l} \exists p_0, \dots, p_M \in E[X]^{<2^n} \text{ s.t.} \\ d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta \\ \wedge \bigwedge_{k=0}^M \langle P_k, L_{\omega, \lambda} \rangle_{H_n} = v_{\lambda}^{(k)} \end{array} \right\}, \quad (45)$$

and its soundness error of which is proven via standard-techniques. We again focus on the list decoding regime, with proximity parameter $\theta = 1 - (1 + \frac{1}{2m}) \cdot \sqrt{\rho}$.

Lemma 5 (Soundness row-batching). *Round 1 of Protocol 3 is a randomized reduction $\mathcal{R} \xrightarrow{\lambda} \mathcal{R}'$ with soundness error*

$$\varepsilon' \leq \ell_m \cdot \frac{d' - 1}{|F|},$$

where $\ell_m = \frac{m + \frac{1}{2}}{\sqrt{\rho}}$ is the Guruswami-Sudan list size bound.

Proof. First of all, note the existence of a single λ so that $(g_0, \dots, g_M) \in \mathcal{R}'$, implies that $(g_0, \dots, g_M)$ is θ -close to $\mathcal{C}^{M+1}$, a Reed-Solomon code over a degree $M + 1$ extension of E , with same evaluation domain D and rate as $\mathcal{C}$. Then, by the Guruswami-Sudan list size bound, there exist at most $\ell_m = (m + \frac{1}{2})/\sqrt{\rho}$ many $(p_0, \dots, p_M) \in \mathcal{P}_n^{M+1}$ so that $d((g_0, \dots, g_M), (p_0, \dots, p_M)) < \theta$, where $\mathcal{P}_n = E[X]^{<2^n}$.

Now assume that for more than $\ell_m \cdot (d' - 1)$ many verifier randomnesses λ , the presumed functions $(g_0, \dots, g_M)$ is in $\mathcal{R}'$. Since the number of θ -proximate polynomials $(p_0, \dots, p_M) \in \mathcal{C}^{M+1}$ is bounded by ℓ_m , we conclude by pigeon-hole principle that at least one of them, denoted again by $(p_0, \dots, p_M)$, satisfies

$$\bigwedge_{k=0}^M \langle P_k, L_{\omega, \lambda} \rangle = v_{\lambda}^{(k)},$$

for at least $(d' - 1)$ different locations λ , where $P_0, \dots, P_M$ are the multivariate representations of $p_0, \dots, p_M$. Both $L_{\omega, \lambda}$ and $v_{\lambda}^{(k)}$ are polynomials in λ of degree at most $d' - 1$, and we conclude from their definitions in (43) that

$$\langle P_k, L_{\omega, j} \rangle = \sum_{i=1}^d \beta_i \cdot v_{i, j}^{(k)},$$

for every $j = 1, \dots, d'$ and $k = 0, \dots, M$. In other words, $\langle P_{k, i}, L_{\omega} \rangle = v_i^{(k)}$, for all i and k . This shows that $(g_0, \dots, g_M)$ belongs to $\mathcal{R}$, proving the theorem. $\square$

We summarize the complete soundness statement in a separate theorem.

Theorem 6 (FRI-Binius soundness). *Protocol 3 is a round-by-round sound interactive oracle proof for the “language” $\mathcal{R}$, with round-by-round soundness error*

$$\varepsilon = \max \{ \varepsilon', \varepsilon_{BF}(\mathcal{C}, M, \theta, s) \}$$

where $\varepsilon_{BF}(\mathcal{C}, M, \theta, s)$ is the round-by-round soundness error of Basefold for inner products (Section 6.1) with $\mathcal{C} = \text{RS}_{2^n}[F, D]$, batch size $M + 1$, proximity parameter $\theta = (1 + \frac{1}{2m}) \cdot \sqrt{\rho}$, where $m \geq 3$, $s \geq 1$ samples, and where $\ell_m = \frac{m + \frac{1}{2}}{\sqrt{\rho}}$ is the Guruswami-Sudan list size bound.

Proof. Follows from Lemma 5 and Section 6.2. $\square$

6.4 WHIR

In overly simple words, WHIR [ACFY24] is a Basefold variant in which the prover commits the foldings with increased redundancy, thereby allowing to detect deviations with fewer samples. Each folded polynomial is committed over a larger domain than just half the previous one, and consistency is ensured similarly to the DEEP method [BGKS20], paraphrasing out-of-domain values as the inner product of the code word with the point evaluation kernel. These inner products are aggregated into the the initial evaluation claim inner product, updating the inner product kernel from round to round.

While [ACFY24] prove the soundness of WHIR in the unique decoding regime, their analysis in the list decoding regime is based on a conjectured correlated agreement property, called *mutual correlated agreement*. Without going into details, mutual correlated agreement is a “global” property, which states that,

except for few folding randomnesses, every proximate of a folding is the folding of a proximate. This conjectured property can be in fact proven by generalizing the list decoder analysis from [BCI⁺20] accordingly, however in this writeup we choose a different path, demonstrating that such a global correlated agreement property is not needed. Instead, the co-curvilinear generalization of the subcode correlated agreement theorem, Theorem 4, is again sufficient to prove soundness, for the same reasons as it helps in the analysis of Basefold on more expressive inner products, Section 6.1.

We demonstrate this on a variation of WHIR, which again follows the values-to-coefficients paradigm as used throughout this write-up. Note that this yields a slightly different folding mechanism as in [ACFY24], and a slightly different point evaluation kernel: The value of

$$p(X) = \sum_{i=0}^{2^n-1} P(i_1, \dots, i_n) \cdot X^i,$$

with $i = i_1 + 2 \cdot i_2 + \dots + 2^{n-1} \cdot i_n$, at a univariate query $z \in F$ corresponds to the inner product

$$p(z) = \sum_{i=0}^{2^n-1} P(i_1, \dots, i_n) \cdot z^{i_1} \cdot z^{2 \cdot i_2} \cdot \dots \cdot z^{2^{n-1} \cdot i_n} = \langle P, Q_n(z) \rangle_{H_n}$$

with the multilinear

$$Q_n(z)(w_1, \dots, w_n) = \prod_{i=1}^n (1 + w_i \cdot (z^{2^{i-1}} - 1)), \quad (46)$$

instead of the n -dimensional Lagrangian $L_n(\mathbf{w}, \mathbf{z})$ at $\mathbf{z} = (z, z^2, \dots, z^{2^{n-1}})$ as in [ACFY24]. The evaluation kernel $Q_n(z)$ shares the same succinct evaluation property, its value at $\mathbf{w} \in F^n$ can be computed withing $O(n)$ field operations.

For the sake of brevity, we restrict our considerations to the first folding round of WHIR for a single multilinear evaluation claim,

$$\langle P, L_n(\cdot, \mathbf{w}) \rangle = v.$$

The other rounds are treated in the same manner. An extension to a fully fledged version with several polynomials, or sumcheck of higher degrees, is not difficult. A round of WHIR is like a Basefold reduction step while omitting $k \geq 2$ commitments for the intermediate foldings. Given

$$g_0 \in F^{D_0},$$

allegedly the word of a polynomial $p_0(X)$ from $\mathcal{P}_n = F[X]^{<2^n}$, the multilinear representation of which, $P(X_1, \dots, X_n) = \Phi^{-1}(p_0)$, satisfies the above inner product claim, the sumcheck protocol is run for k steps, but without committing to the intermediate foldings

$$\text{fold}_{\lambda_1}(p_0), \text{fold}_{\lambda_2}(\text{fold}_{\lambda_1}(p_0)), \dots,$$

where $\lambda_1, \lambda_2, \dots$ are the verifier randomnesses. Only the final k -th folding $p_k(X) \in \mathcal{P}_{n-k}$ is committed, as word

$$g_1 \in F^{D_1}$$

over a domain D_1 larger and (for technical reasons) disjoint to the image $\pi^k(D_0)$,

$$\pi^k(D_0) \cap D_1 = \emptyset.$$

Consistency between g_0 and g_1 is then tested in a separate query phase for this round, which uses the first oracle g_0 to derive the values of the folding at the out-of-domain points from $\pi^k(D_0)$. For soundness reasons, the folding behind f_1 is queried at a further out-of-domain point, drawn from the much larger sample space $F \setminus D_1$. These evaluation claims are expressed as inner products over the hypercube H_{n-k} , and combined into a single claim in a separate aggregation step.

Protocol 4 (folding round of WHIR). *Given $f_0 \in F^{D_0}$, allegedly the values of a polynomial $p_0 \in \mathcal{P}_n$ the multilinear representation of which satisfies the*

$$\langle P_0, R_0 \rangle_{H_n} = c_0$$

for some public and succinctly evaluable multilinear $R_0 \in F[X_1, \dots, X_n]$.

1. *Sumcheck. Both prover and verifier run the sumcheck for $k \geq 2$ steps. In the course of it, the prover commits to the quadratic sumcheck polynomials $q_0(X), \dots, q_{k-1}(X)$, and the verifier has sampled $\lambda_1, \dots, \lambda_k \leftarrow_{\$} F$. The prover commits to the k -th folding*

$$p_k = \text{fold}_{\lambda_1, \dots, \lambda_k}(p_0) \in \mathcal{P}_{n-k}$$

over the domain D_1 , where the k -th folding operator is defined as in (32).

2. *Query phase. The verifier chooses an out-of-domain query $z_0 \leftarrow_{\$} F \setminus D_1$ on which the prover responds with the value*

$$v_0 = p_k(z_0).$$

The verifier further samples $z_1, \dots, z_{s_0} \leftarrow_{\$} \pi^k(D_0)$, and uses f_0 to determine the expected values

$$v_j = \text{fold}_{\lambda_1, \dots, \lambda_k}(f_0)(z_j), \quad j = 1, \dots, s_0.$$

3. *Aggregation. The verifier samples $\gamma \leftarrow_{\$} F$ for combining all the evaluation claims with the specialization of the sumcheck.*

$$\begin{aligned} \langle P_k, R_0(\lambda_1, \dots, \lambda_k, \cdot) \rangle_{H_{n-k}} &= q_{k-1}(\lambda_k), \\ \langle P_k, Q_{n-k}(z_j) \rangle_{H_{n-k}} &= v_j, \quad j = 0, \dots, s_0, \end{aligned}$$

into a the single inner product claim

$$\langle P_k, R_1 \rangle_{H_{n-k}} = c_1$$

with the succinctly evaluable multilinear $R_1 \in F[X_{k+1}, \dots, X_n]$ defined as

$$R_1 = R_0(\lambda_1, \dots, \lambda_k, \cdot) + \sum_{j=0}^{s_0} \gamma^{j+1} \cdot Q_{n-k}(z_j), \quad (47)$$

and $c_1 = q_{k-1}(\lambda_k) + \sum_{j=0}^{s_0} \gamma^{j+1} \cdot v_j$.

The analysis of Protocol 4 similar to that of Basefold with general inner product kernels, Section 6.1. Given proximity parameter θ_0 , the target “language” is

$$\mathcal{R}_0 = \left\{ f_0 \in F^{D_0} : \begin{array}{l} \exists p_0(X) \in \mathcal{P}_n, d_{D_0}(f_0, p_0) \leq \theta_0, \\ \langle P_0, R_0 \rangle_{H_n} = c_0 \end{array} \right\},$$

and each step of the sumcheck is a randomized reduction $\mathcal{R}_{i-1} \xrightarrow{\lambda_i} \mathcal{R}_i$, where

$$\mathcal{R}_i = \left\{ \begin{array}{l} q_{j-1}(\lambda_j) = q_j(0) + q_j(1), j = 1, \dots, i-1, \\ \text{and} \\ f_0 \in F^{D_0}, \\ q_0, \dots, q_{i-1} \in F[X]^{\leq 2} : \end{array} \begin{array}{l} \exists p_i(X) \in \mathcal{P}_{n-i}, \\ d_{\pi^i(D_0)}(\text{fold}_{\lambda_1, \dots, \lambda_i}(f_0), p_i) \leq \theta_0, \\ \langle P_i, R_0(\lambda_1, \dots, \lambda_i, \cdot) \rangle = q_{i-1}(\lambda_i), \end{array} \right\},$$

for $i = 1, \dots, k-1$. In the last sumcheck step, where the prover also sends also a proposal $f_k \in F^{D_1}$ for the k -th folding, the relation is stricter,

$$\mathcal{R}_k = \left\{ \begin{array}{l} q_{j-1}(\lambda_j) = q_j(0) + q_j(1), j = 1, \dots, k-1, \\ \text{and} \\ f_0 \in F^{D_0}, f_k \in F^{D_1}, \\ q_0, \dots, q_{k-1} \in F[X]^{\leq 2} : \end{array} \begin{array}{l} \exists p_k(X) \in \mathcal{P}_{n-k}, d_{D_1}(f_k, p_k) \leq \theta_1, \\ d_{\pi^k(D_0)}(\text{fold}_{\lambda_1, \dots, \lambda_k}(f_0), p_k) \leq \theta_0, \\ \langle P_k, R_0(\lambda_1, \dots, \lambda_k, \cdot) \rangle = q_{k-1}(\lambda_k), \end{array} \right\},$$

where θ_1 is another (typically larger) proximity parameter, leveraging the increased redundancy over D_1 . The query phase is a reduction $\mathcal{R}_k \xrightarrow{z_0, z_1, \dots, z_{s_0}} \mathcal{R}_{k+1}$, where

$$\mathcal{R}_{k+1} = \left\{ \begin{array}{l} q_{j-1}(\lambda_j) = q_j(0) + q_j(1), j = 1, \dots, k-1, \\ v_j = \text{fold}_{\lambda_1, \dots, \lambda_k}(f_0)(z_j), j = 1, \dots, s_0, \\ \text{and} \\ f_0 \in F^{D_0}, f_k \in F^{D_1}, \\ q_0, \dots, q_{k-1} \in F[X]^{\leq 2} : \\ v_0 \end{array} \begin{array}{l} \exists p_k(X) \in \mathcal{P}_{n-k}, d_{D_1}(f_k, p_k) \leq \theta_1, \\ \langle P_k, R_0(\lambda_1, \dots, \lambda_k, \cdot) \rangle = q_{k-1}(\lambda_k), \\ \langle P_k, Q_{n-k}(z_j) \rangle = v_j, j = 0, \dots, s_0. \end{array} \right\}.$$

The DEEP query z_0 is for narrowing down the proximates to a single one, and the remaining in-domain queries are used to test its distance to the folding. Eventually, the aggregation step $\mathcal{R}_{k+1} \xrightarrow{\gamma} \mathcal{R}_{k+2}$ reduces to

$$\mathcal{R}_{k+2} = \left\{ \begin{array}{l} q_{j-1}(\lambda_j) = q_j(0) + q_j(1), j = 1, \dots, k-1, \\ \text{fold}_{\lambda_1, \dots, \lambda_k}(f_0)(z_j) = v_j, j = 1, \dots, s_0, \\ \text{and} \\ f_0 \in F^{D_0}, f_k \in F^{D_1}, \\ q_0, \dots, q_{k-1} \in F[X]^{\leq 2} : \\ v_0 \end{array} \begin{array}{l} \exists p_k(X) \in \mathcal{P}_{n-k}, d_{D_1}(f_k, p_k) \leq \theta_1, \\ \langle P_k, R_1 \rangle_{H_{n-k}} = c_1 \end{array} \right\},$$

with the combined multilinear as above. We split the analysis into three lemmas.

Lemma 6 (Soundness sumcheck rounds). *Each round in Step 1 of Protocol 4, $1 \leq i \leq k$, is a randomized reduction $\mathcal{R}_{i-1} \xrightarrow{\lambda_i} \mathcal{R}_i$ with soundness error*

$$\varepsilon_i = \varepsilon(\mathcal{C}_i, 1, 1, \theta_0).$$

Here, $\varepsilon(\mathcal{C}_i, 1, 1, \theta_0)$ is the error from Theorem 4, for the Reed-Solomon code $\mathcal{C}_i = \text{RS}(F, \pi^i(D_0), \rho_0)$ generated by $\mathcal{P}_{n-i}$, with $M = 1$ and trivial weights $B = 1$.

Proof. For the steps $i = 1, \dots, k-1$, the proof is verbatim to that of Lemma 4, using trivial consistency weights. (As in Section 5.4, one can think of the intermediate foldings as implicitly committed with f_0 .) The same applies to the last reduction step $i = k$, in which the prover is required to commit to $f_k \in F^{D_1}$, which has a θ_1 -proximate $p_k \in \mathcal{P}_{n-k}$ which moreover is a θ_0 -proximate of the folding $\text{fold}_{\lambda_1, \dots, \lambda_k}(f_0)$ over the projected domain $\pi^k(D_0)$. $\square$

Lemma 7 (Soundness query phase). *Step 2 of Protocol 4 is a randomized reduction $\mathcal{R}_k \xrightarrow{z_0, z_1, \dots, z_{s_0}} \mathcal{R}_{k+1}$ with soundness error*

$$\varepsilon_{k+1} = \frac{\ell_1(\theta_1)^2}{2} \cdot \frac{2^{n-k}}{|F \setminus D_1|} + (1 - \theta_0)^{s_0},$$

where $\ell_1(\theta_1)$ is the number of θ_1 -proximates for f_1 , i.e. the GS list size bound of the Reed-Solomon code generated by $\mathcal{P}_{n-k}$ over D_1 .

Proof. The first term in the claimed error is the probability that there is more than one proximate with the same value at $z_0 \leftarrow F \setminus D_1$. Except with this probability, there is exactly one proximate p_k which evaluates to v_0 at z_0 . The second term stands for the probability that over $\pi^k(D_0)$, this unique proximate has distance to $\text{fold}_{\lambda_1, \dots, \lambda_k}(f_0)$ greater than θ_0 but is not detected by the samples $z_1, \dots, z_{s_0}$ drawn from $\pi^k(D_0)$. $\square$

Lemma 8 (Soundness aggregation). *Step 3 of Protocol 4 is a randomized reduction $\mathcal{R}_{k+1} \xrightarrow{\gamma} \mathcal{R}_{k+2}$ with soundness error*

$$\varepsilon_{k+2} = \ell_1(\theta_1) \cdot \frac{s_0}{|F|},$$

with $\ell_1(\theta_1)$ as above.

Proof. This is the usual pigeon-hole argument on the θ_1 -proximates of f_k , which are at most $\ell_1(\theta_1)$ many. If there are more than $\ell_1(\theta_1) \cdot s_0$ many γ , for which there exists a proximate as claimed in $\mathcal{R}_{k+2}$, then at least one them satisfies the combined claim holding for more than s_0 such γ . By the degree in γ , this proximate satisfies the component claims. $\square$

References

- ACFY24. Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. WHIR: Reed-solomon proximity testing with super-fast verification. In *IACR preprint archive 2024/1586*, 2024. <https://eprint.iacr.org/2024/1586>.
- AFLN23. Martin R. Albrecht, Giacomo Fenzi, Oleksandra Lapiha, and Ngoc Khanh Nguyen. SLAP: Succinct lattice-based polynomial commitments from standard assumptions. 2023. <https://eprint.iacr.org/2023/1469>.
- AHIV17. Scott Ames, Carmit Hazay, Yuval Ishai, and Muthuramakrishnan Venkatasubramanian. Liger: Lightweight sublinear arguments without a trusted setup. In *CCS 2017*, 2017. Full paper: <https://eprint.iacr.org/2022/1608>.
- BBB⁺18. Benedikt Bünz, Jonathan Bootle, Dan Boneh, Andrew Poelstra, Pieter Wuille, and Greg Maxwell. Bulletproofs: Short proofs for confidential transactions and more. In *in IEEE Symposium on Security and Privacy*, pages 315–334, 2018. Full paper: <https://eprint.iacr.org/2017/1066>.
- BBHR18. Eli Ben-Sasson, Iddo Bentov, Yinon Horesh, and Michael Riabzev. Fast Reed-Solomon interactive oracle proofs of proximity. In *ICALP 2018*, 2018. Full paper: <https://eccc.weizmann.ac.il/report/2017/134/>.
- BCC⁺16. Jonathan Bootle, Andrea Cerulli, Pyrros Chaidos, Jens Groth, and Christophe Petit. Efficient zero-knowledge arguments for arithmetic circuits in the discrete log setting. In M. Fischlin and J.S. Coron, editors, *EUROCRYPT 2016*, volume 9666 of *LNCS*. Springer, 2016. Full paper: <https://eprint.iacr.org/2016/263>.
- BCHO22. Jonathan Bootle, Alessandro Chiesa, Yuncong Hu, and Michele Orrù. Gemini: Elastic SNARKs for diverse environments. In *EUROCRYPT 2022*, 2022. Full paper: <https://eprint.iacr.org/2022/420>.
- BCI⁺20. Eli Ben-Sasson, Dan Carmon, Yuval Ishai, Swastik Kopparty, and Shubhangi Saraf. Proximity gaps for Reed-Solomon codes. In *FOCS 2020*, 2020. Full paper: <https://eprint.iacr.org/2020/654>.
- BCKL21. Eli Ben-Sasson, Dan Carmon, Swastik Kopparty, and David Levit. Elliptic Curve Fast Fourier Transform (ECFFT) Part I: Fast polynomial algorithms over all finite fields. In *Electronic Colloquium on Computational Complexity*, volume TR21-103, 2021. <https://eccc.weizmann.ac.il/report/2021/103/>.
- BCKL22. Eli Ben-Sasson, Dan Carmon, Swastik Kopparty, and David Levit. Scalable and transparent proofs over all large fields, via elliptic curves (ECFFT part II). In *IACR preprint archive*, 2022. <https://eprint.iacr.org/2022/1542>.
- BCS16. Eli Ben-Sasson, Alessandro Chiesa, and Nicholas Spooner. Interactive oracle proofs. In *TCC 2016*, pages 31–60, 2016.
- BFS20. Benedikt Bünz, Ben Fisch, and Alan Szepieniec. Transparent SNARKs from DARK compilers. In *EUROCRYPT 2020*, 2020. Full paper: <https://eprint.iacr.org/2019/1229>.
- BGKS20. Eli Ben-Sasson, Lior Goldberg, Swastik Kopparty, and Shubhangi Saraf. DEEP-FRI: Sampling outside the box improves soundness. In *ITCS 2020*, 2020. Full paper: <https://eprint.iacr.org/2019/336>.
- BLNR22. Sarah Bordage, Mathieu Lhotel, Jade Nardi, and Hugues Randriam. Interactive oracle proofs of proximity to algebraic geometry codes. In *CCS’22*, 2022. Full paper: <https://arxiv.org/abs/2011.04295>.

- BM88. Lázló Babai and Shlomer Moran. Arthur-Merlin games: A randomized proof system and a hierarchy of complexity classes. In *Journal of Computer Sciences*, 1988.
- BS84. Lázló Babai and Endre Szemeredi. On the complexity classes of matrix group problems. In *Proc. of the 25th Symposium on Foundation of Computer Sciences*, 1984. <https://doi.org/10.1109/SFCS.1984.71591>.
- CBBZ22. Binyi Chen, Benedikt Bünz, Dan Boneh, and Zhenfei Zhang. Hyperplonk: PLONK with a linear-time prover and high-degree costumed gates. In *IACR ePrint Archive 2020/1355*, 2022. <https://eprint.iacr.org/2022/1355>.
- CCH⁺18. Ran Canetti, Yilei Chen, Justin Holmgren, Alex Lombardi, Guy N. Rothblum, and Ron D. Rothblum. Fiat-shamir from simpler assumptions. 2018. <https://eprint.iacr.org/2018/1004>.
- CMNW24. Valerio Cini, Giulio Malavolta, Ngoc Khanh Nguyen, and Hoeteck Wee. Polynomial commitments from lattices: Post-quantum security, fast verification and transparent setup. 2024. <https://eprint.iacr.org/2024/281>.
- CMS19. Alessandro Chiesa, Peter Manohar, and Nicholas Spooner. Succinct arguments in the quantum random oracle model. In *TCC 2019*, pages 1 – 29, 2019. Full paper: <https://eprint.iacr.org/2019/834>.
- CY24. Alessandro Chiesa and Eylon Yogev. Building cryptographic proofs from hash functions. 2024. <https://snargsbook.org/>.
- Dia24a. Benjamin E. Diamond. Galois FRI-Binius: Joseph Jonston’s Idea. [hackmd.io](https://hackmd.io/@benediamond/BkLYpf__0), 2024. https://hackmd.io/@benediamond/BkLYpf__0.
- Dia24b. Benjamin E. Diamond. Overhead-Free FRI-Binius Without Galois Theory. [hackmd.io](https://hackmd.io/@benediamond/BJgKxUau0), 2024. <https://hackmd.io/@benediamond/BJgKxUau0>.
- DP23. Benjamin E. Diamond and Jim Posen. Succinct arguments over towers of binary fields. In *IACR preprint archive 2023/1784*, 2023. <https://eprint.iacr.org/2023/1784>.
- DP24. Benjamin E. Diamond and Jim Posen. Polylogarithmic proofs for multilinear over binary towers. In *IACR preprint archive 2024/504*, 2024. <https://eprint.iacr.org/2024/504>.
- DT24. Quand Dao and Justin Thaler. More optimizations to sum-check proving. In *IACR ePrint Archive 2024/1210*, 2024. <https://eprint.iacr.org/2024/1210>.
- GM10. Shuhong Gao and Todd Mateer. Additive Fast Fourier Transforms over finite fields. In *IEEE Transactions on Information Theory*, volume 56 (12), 2010.
- GMR89. Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. In *SIAM J. Comput.*, 1989.
- Gru24. Angus Gruen. Some improvements for the PIOP for ZeroCheck. In *IACR ePrint Archive 2024/108*, 2024. <https://eprint.iacr.org/2024/108>.
- Hab24. Ulrich Haböck. Basefold in the list decoding regime. In *IACR preprint archive 2024/1586*, 2024. <https://eprint.iacr.org/archive/2024/1571/20241005:084637>.
- Ham22. Adrian Hamelink. Gemini. [hackmd.io](https://hackmd.io/@adrian-aztec/BJxoyeCqj), 2022. <https://hackmd.io/@adrian-aztec/BJxoyeCqj>.
- HLP24. Ulrich Haböck, David Levit, and Shahar Papini. Circle STARKs. In *IACR preprint archive*, 2024. <https://eprint.iacr.org/2024/278>.
- KT23. Tohru Kohrita and Patrick Towa. Zeromorph: Zero-knowledge multilinear-evaluation proofs from homomorphic univariate commitments. In *IACR preprint archive 2023/917*, 2023. <https://eprint.iacr.org/2023/917>.

- KZG10. Aniket Kate, Gregory M. Zaverucha, and Ian Goldberg. Constant-size commitments to polynomials and their applications. In Abe M., editor, *ASIACRYPT 2010*, volume 6477 of *LNCS*. Springer, 2010.
- LCH14. Sian-Jheng Lin, Wei-Ho Chung, and Yung-Hsiang S. Han. Novel polynomial basis and its application to Reed-Solomon erasure codes. In *FOCS 2014*, 2014.
- LFKN92. Carsten Lund, Lance Fortnow, Howard Karloff, and Noam Nisan. Algebraic methods for interactive proof systems. In *Journal of the Association for Computing Machinery*, volume 39, pages 859–868, 1992. Full paper: <https://eprint.iacr.org/2017/1066>.
- NS24. Ngoc Khanh Nguyen and Gregor Seiler. Greyhound: Fast polynomial commitments from lattices. 2024. <https://eprint.iacr.org/2024/1293>.
- PH23. Shahar Papini and Ulrich Haböck. Improving logarithmic derivative lookups using GKR. In *IACR ePrint Archive 2023/1284*, 2023. <https://eprint.iacr.org/2023/1284>.
- PST13. Charalampos Papamanthou, Elaine Shi, and Roberto Tamassia. Signatures of correct computation. In *TCC 3*, volume 7785 of *LNCS*. Springer, 2013.
- RRR16. Omer Reingold, Ron Rothblum, and Guy Rothblum. Constant-round interactive proofs for delegating computation. In *STOC'16*, 2016. Full paper: <https://eccc.weizmann.ac.il/report/2016/061/>.
- SB23. István András Seres and Péter Burcsi. Behemoth: transparent polynomial commitment scheme with constant opening proof size and verifier time. 2023. <https://eprint.iacr.org/2023/670>.
- Tha23. Justin Thaler. Proofs, arguments, and zero-knowledge. 2023. <https://people.cs.georgetown.edu/jthaler/ProofsArgsAndZK.html>.
- ZCF23. Hadas Zeilberger, Binyi Chen, and Ben Fisch. BaseFold: efficient field-agnostic polynomial commitment schemes from foldable codes. In *IACR preprint archive 2023/1705*, 2023. <https://eprint.iacr.org/2023/1705>.
- ZXZS20. Jiaheng Zhang, Tianchen Xie, Yupeng Zhang, and Dawn Song. Transparent polynomial delegation and its applications to zero knowledge proof. In *IEEE S&P 2020*, 2020. Full paper: <https://eprint.iacr.org/2019/1482>.